Geometry 3

From an arbitrary parameter t to arc length s (and curvature definitions)

1) Reparametrization by arc length

Let $\alpha : I \to \mathbb{R}^2$ be a regular planar curve

$$\alpha(t) = (f(t), g(t)), \qquad \alpha'(t) \neq 0.$$

Define the *arc length function* (measured from a fixed $t_0 \in I$) by

$$s(t) = \int_{t_0}^{t} \|\alpha'(u)\| \, du = \int_{t_0}^{t} \sqrt{(f'(u))^2 + (g'(u))^2} \, du.$$

Then $s'(t) = \|\alpha'(t)\| > 0$, so $s(t)$ is strictly increasing and has a local inverse $t = t(s)$. The *unit-speed (arc-length) parametrization* is

$$\beta(s) = \alpha(t(s)).$$

It satisfies $\|\beta'(s)\| = 1$.

2) Curvature of the generating curve

For a unit-speed planar curve $\beta(s) = (x(s), y(s))$ the (signed) curvature is

$$\kappa(s) = x'(s)y''(s) - y'(s)x''(s),$$

and the (unsigned) curvature is $|\kappa(s)| = \|\beta''(s)\|$.

If $\beta(s) = (f(s), g(s))$ is unit-speed, then

$$\kappa(s) = f'(s)g''(s) - g'(s)f''(s).$$

3) Curvatures on the surface (normal, principal, mean, Gaussian)

Let S be a surface and $p \in S$ with unit normal N. The *normal curvature* in a tangent direction $t \in T_pS$ is

$$k_n(t) = \frac{\mathrm{II}(t,t)}{\mathrm{I}(t,t)} \quad (t \neq 0),$$

and for $\|t\| = 1$ simply $k_n(t) = \mathrm{II}(t,t)$. The *principal directions* are the eigen-directions of the shape operator

$$S_p(v) = -dN_p(v),$$

and the corresponding eigenvalues k_1, k_2 are the *principal curvatures*. The *mean curvature* and *Gaussian curvature* are

$$H = \frac{k_1 + k_2}{2}, \qquad K = k_1 k_2.$$

In a principal orthonormal basis, Euler's formula gives

$$k_n(\theta) = k_1 \cos^2\theta + k_2 \sin^2\theta.$$

4) Link to Problem 8 (surface of revolution)

For the surface of revolution

$$\phi(u, s) = \big(f(s)\cos u,\ f(s)\sin u,\ g(s)\big),$$

where the generating curve $(f(s), g(s))$ is unit-speed, i.e. $(f')^2 + (g')^2 = 1$, the first fundamental form satisfies $E = f^2$, $F = 0$, $G = 1$ and the principal curvatures are

$$k_1 = -\frac{g'}{f}, \qquad k_2 = f''g' - f'g''.$$

Moreover, since for a unit-speed planar curve $\kappa = f'g'' - g'f''$, we have

$$k_2 = -(f'g'' - g'f'') = -\kappa,$$

(up to sign depending on the chosen orientation of the surface normal).

Example: a cone from a non-unit-speed generating curve

Take a straight line (not necessarily unit-speed)

$$\alpha(t) = (at, bt), \qquad a > 0,\ b \in \mathbb{R},$$

which, when revolved around the z-axis, generates a (right circular) cone (away from the apex).

Its speed is constant:

$$\|\alpha'(t)\| = \sqrt{a^2 + b^2}.$$

Hence the arc length is

$$s(t) = \int_0^t \sqrt{a^2 + b^2}\, du = t\sqrt{a^2 + b^2}, \qquad t = \frac{s}{\sqrt{a^2 + b^2}}.$$

So the unit-speed profile is

$$f(s) = a\,\frac{s}{\sqrt{a^2 + b^2}}, \qquad g(s) = b\,\frac{s}{\sqrt{a^2 + b^2}},$$

and

$$f' = \frac{a}{\sqrt{a^2 + b^2}}, \quad g' = \frac{b}{\sqrt{a^2 + b^2}}, \quad f'' = g'' = 0.$$

Therefore

$$k_1 = -\frac{g'}{f} = -\frac{\frac{b}{\sqrt{a^2+b^2}}}{\frac{a}{\sqrt{a^2+b^2}}\, s} = -\frac{b}{a}\,\frac{1}{s}, \qquad k_2 = f''g' - f'g'' = 0.$$

So

$$K = k_1 k_2 = 0, \qquad H = \frac{k_1 + k_2}{2} = -\frac{b}{2a}\,\frac{1}{s},$$

(with the sign depending on the chosen orientation). Geometrically, one principal curvature vanishes because a cone is *developable* (it can be unrolled onto a plane).

Invertibility of $s(t)$ and recovering the curve

Yes — locally (and in many examples, globally) one can invert $s(t)$ and recover the curve as a function of the arc-length parameter s, provided the curve is regular (i.e. $\|\alpha'(t)\| \neq 0$) and $s(t)$ is defined in a monotone way.

General fact. Let $\alpha : I \to \mathbb{R}^2$ be a regular curve. Define the arc length (from a fixed $t_0 \in I$) by

$$s(t) = \int_{t_0}^{t} \|\alpha'(u)\| \, du.$$

Then

$$s'(t) = \|\alpha'(t)\| > 0,$$

so $s(t)$ is strictly increasing and therefore has a local inverse $t = t(s)$. The arc-length parametrization of the same geometric curve is then

$$\beta(s) = \alpha\big(t(s)\big),$$

and it satisfies $\|\beta'(s)\| = 1$.

Line example $\alpha(t) = (at, bt)$. Here

$$\alpha'(t) = (a, b), \qquad \|\alpha'(t)\| = \sqrt{a^2 + b^2},$$

so (taking $t_0 = 0$)

$$s(t) = \int_0^t \sqrt{a^2 + b^2} \, du = t\sqrt{a^2 + b^2}.$$

This is invertible for all t (global inverse), namely

$$t(s) = \frac{s}{\sqrt{a^2 + b^2}}.$$

Hence the curve as a function of arc length is

$$\beta(s) = \alpha\big(t(s)\big) = \left(a \, \frac{s}{\sqrt{a^2 + b^2}}, \; b \, \frac{s}{\sqrt{a^2 + b^2}} \right).$$

Small caveat. One does not usually recover the *original* parameter t uniquely if one allows shifts (different choices of t_0) or reversed orientation, but one does recover the same geometric curve together with an arc-length parametrization of it.

Problem 9. Gauss map: surjectivity, parity, and an elliptic point

Let S be a compact orientable surface in $\mathbb{R}^3$. Show that the Gauss map is surjective and that it hits almost every unit vector the same number of times modulo 2. (You may use the Jordan–Brouwer separation theorem.) Show that S always has an elliptic point.

Theory

For an oriented surface $S \subset \mathbb{R}^3$, the *Gauss map* is

$$N : S \to S^2, \qquad p \mapsto N(p),$$

where $N(p)$ is the unit normal at p. For $a \in S^2$, define the *height function*

$$h_a(p) = \langle p, a \rangle.$$

Since S is compact, h_a attains a maximum and a minimum. At a critical point p of h_a, the tangential component of a vanishes, hence a is normal to S at p, i.e. $N(p) = \pm a$.

By Sard's theorem, the set of regular values of N has full measure in S^2; for a regular value y, the preimage $N^{-1}(y)$ is finite. For maps between compact 2-manifolds, the parity $\#N^{-1}(y) \bmod 2$ is constant on the set of regular values (the mod-2 degree).

Solution

(1) Surjectivity. Fix $a \in S^2$. The function $h_a(p) = \langle p, a \rangle$ attains a maximum at some $p_{\max} \in S$ and a minimum at some $p_{\min} \in S$. At each extremum, a is normal to S, hence

$$N(p_{\max}) = \pm a, \qquad N(p_{\min}) = \mp a.$$

Thus a lies in the image of N, and since a was arbitrary, N is surjective.

(2) Parity of the number of preimages. Let $y_0, y_1 \in S^2$ be regular values of N. Choose a smooth path $\gamma : [0,1] \to S^2$ from y_0 to y_1 avoiding the critical values of N. Then $M := N^{-1}(\gamma([0,1]))$ is a compact 1-manifold with boundary

$$\partial M = N^{-1}(y_0) \cup N^{-1}(y_1).$$

A compact 1-manifold has an even number of boundary points, hence

$$\#N^{-1}(y_0) \equiv \#N^{-1}(y_1) \pmod 2.$$

Therefore, for almost every $y \in S^2$ (namely for every regular value) the number $\#N^{-1}(y)$ is constant modulo 2.

(3) Existence of an elliptic point. Choose $a \in S^2$ to be a regular value of N (possible since regular values have full measure). Let p be a maximum point of h_a on S. Then $N(p) = \pm a$. Since a is a regular value, dN_p is invertible, so $\det(dN_p) \neq 0$. At a (nondegenerate) maximum the second fundamental form is definite, hence the principal curvatures have the same sign and are nonzero. Consequently $K(p) > 0$, i.e. p is an elliptic point.

Examples and intuition

- On a sphere, N is (up to orientation) a diffeomorphism, so every direction is hit exactly once.
- On a torus in $\mathbb{R}^3$, most directions are hit several times; nevertheless the parity $\#N^{-1}(y) \bmod 2$ is constant for regular values y.

- Elliptic points occur at strictly convex supporting points (e.g. the "outermost" points for a generic height direction).

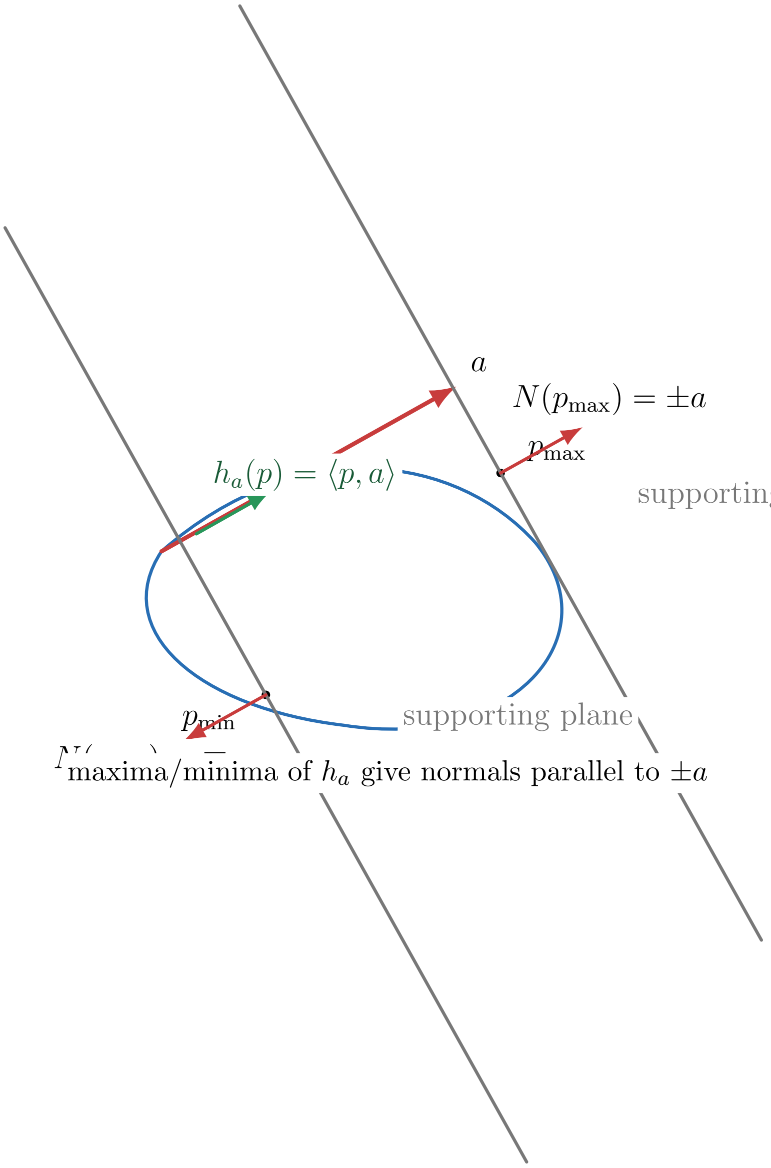


Figure 1: Problem 9: For a fixed direction $a \in S^2$, the height function $h_a(p) = \langle p, a\rangle$ attains a maximum and minimum on compact S. At such supporting points the tangent plane is orthogonal to a, hence $N(p) = \pm a$. This gives surjectivity of the Gauss map.

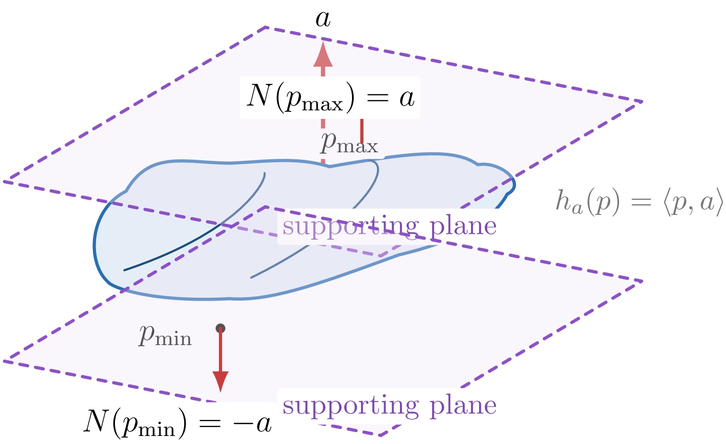


Figure 2: Problem 9 (3D schematic): for a fixed direction a (drawn as e_z), the height function $h_a(p) = \langle p, a\rangle$ has maxima/minima on compact S. The supporting planes $z = \text{const}$ touch S at $p_{\max}, p_{\min}$, where the normals align with $\pm a$.

Problem 10. Totally umbilic compact surfaces are planes or spheres

Show that if S is a connected surface in $\mathbb{R}^3$ such that every point is umbilic, then S is contained in a plane or a sphere. *(Hint: Use that in a parametrization $\phi(u, v)$ one has $N_{uv} = N_{vu}$.*

Theory

A surface is *totally umbilic* if at every point the shape operator is a scalar:

$$S_p = \lambda(p)\, I.$$

Equivalently, the principal curvatures satisfy $k_1 = k_2 = \lambda$, or $\mathrm{II} = \lambda\, \mathrm{I}$. In local coordinates $\phi(u, v)$ with unit normal N, the Weingarten equations give

$$N_u = -S(\phi_u), \qquad N_v = -S(\phi_v).$$

Solution

Assume S is totally umbilic, so $S = \lambda I$. Then

$$N_u = -\lambda\, \phi_u, \qquad N_v = -\lambda\, \phi_v.$$

Differentiate:

$$N_{uv} = -(\lambda_v \phi_u + \lambda \phi_{uv}), \qquad N_{vu} = -(\lambda_u \phi_v + \lambda \phi_{vu}).$$

Since mixed partial derivatives commute, $N_{uv} = N_{vu}$ and $\phi_{uv} = \phi_{vu}$, hence

$$\lambda_v \phi_u = \lambda_u \phi_v.$$

Because ϕ_u and ϕ_v are linearly independent, it follows that

$$\lambda_u = \lambda_v = 0,$$

so λ is constant on each connected component; since S is connected, λ is constant.

If $\lambda = 0$, then $N_u = N_v = 0$, so N is constant and S lies in a plane orthogonal to N.

If $\lambda \neq 0$, define

$$c = \phi + \frac{1}{\lambda} N.$$

Then

$$c_u = \phi_u + \frac{1}{\lambda} N_u = \phi_u - \phi_u = 0, \qquad c_v = \phi_v + \frac{1}{\lambda} N_v = \phi_v - \phi_v = 0,$$

so c is constant. Therefore $\|\phi - c\| = 1/|\lambda|$, i.e. ϕ lies on a sphere of radius $1/|\lambda|$ centered at c.

Examples

- A plane has $k_1 = k_2 = 0$ everywhere (totally umbilic).
- A sphere of radius R has $k_1 = k_2 = \pm 1/R$ everywhere (totally umbilic).

Problem 11. Gauss curvature as spherical area distortion

Let p be a point of a surface S such that the Gaussian curvature $K(p) \neq 0$, and let V be a small connected neighbourhood of p where K does not change sign. Define the *spherical area* $A_N(B)$ of a domain $B \subset V$ to be the area of $N(B)$ if $K(p) > 0$, or minus the area of $N(B)$ if $K(p) < 0$ (where N is the Gauss map). Show that

$$K(p) = \lim_{A(B) \to 0} \frac{A_N(B)}{A(B)},$$

where $A(B)$ is the area of B and the limit is taken through a sequence of domains B_n converging to p (eventually contained in every ball around p).

Theory (area change under the Gauss map)

For an oriented surface, the Gauss map $N : S \to S^2$ has differential $dN_p : T_pS \to T_{N(p)}S^2$. Its Jacobian determinant equals the Gaussian curvature:

$$\det(dN_p) = K(p).$$

If N is a local diffeomorphism on a region B and K has constant sign on B, then the area formula for smooth maps between surfaces yields

$$\text{area}(N(B)) = \int_B |\det(dN_q)| \, dA(q) = \int_B |K(q)| \, dA(q).$$

The signed spherical area $A_N(B)$ is defined precisely so that the absolute value is removed when K has constant sign.

Solution

Solution. Since $K(p) \neq 0$ and K is continuous, after shrinking V we may assume $K(q) \neq 0$ on V, so dN_q is invertible and hence N is a local diffeomorphism on V. Let $B \subset V$ be a domain. By the area formula,

$$\operatorname{area}(N(B)) = \int_B |K(q)| \, dA(q).$$

Because K does not change sign on V, the definition of $A_N(B)$ gives

$$A_N(B) = \int_B K(q) \, dA(q).$$

Therefore

$$\frac{A_N(B)}{A(B)} = \frac{1}{A(B)} \int_B K(q) \, dA(q),$$

which is the average value of K over B. Let $B_n \to p$ with $A(B_n) \to 0$. By continuity of K, the average value over B_n tends to $K(p)$, i.e.

$$\lim_{n\to\infty} \frac{1}{A(B_n)} \int_{B_n} K \, dA = K(p).$$

Hence

$$K(p) = \lim_{A(B)\to 0} \frac{A_N(B)}{A(B)}.$$

Examples and intuition

- **Sphere of radius R.** One has $K \equiv 1/R^2$ and small areas satisfy $\operatorname{area}(N(B)) \sim (1/R^2)\operatorname{area}(B)$, so the ratio tends to $1/R^2$.
- **Plane.** The Gauss map is constant, so $\operatorname{area}(N(B)) = 0$ and the ratio is 0, consistent with $K \equiv 0$.
- **Cylinder.** The Gauss map image is 1-dimensional (a circle), so spherical area is 0 and the ratio is 0, consistent with $K \equiv 0$.

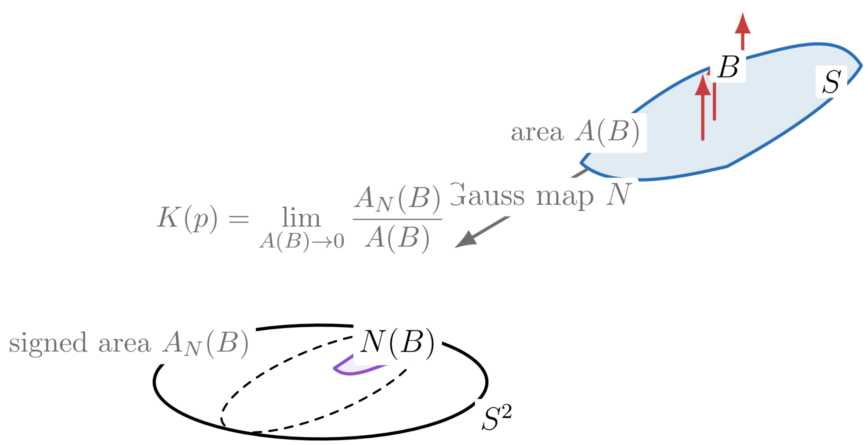


Figure 3: Problem 11: a small patch $B \subset S$ is mapped by the Gauss map N to a patch $N(B) \subset S^2$. The Gaussian curvature at p is the infinitesimal ratio of signed spherical area to surface area.

Why one may view dN_p as a map $T_pS \to T_pS$

Formally, the Gauss map is

$$N : S \to S^2, \qquad p \mapsto N(p),$$

so its differential at p is a linear map

$$dN_p : T_pS \longrightarrow T_{N(p)}S^2.$$

At first glance the domain and codomain are different tangent planes. However, for the Gauss map there is a canonical identification of these two planes inside $\mathbb{R}^3$.

Step 1: $dN_p(v)$ is tangent to the sphere. Since $\|N\|^2 = \langle N, N\rangle = 1$ on S, differentiating at p in the direction $v \in T_pS$ gives

$$0 = d(\langle N, N\rangle)_p(v) = 2\langle dN_p(v), N(p)\rangle.$$

Hence

$$\langle dN_p(v), N(p)\rangle = 0,$$

so $dN_p(v)$ is orthogonal to $N(p)$, i.e.

$$dN_p(v) \in T_{N(p)}S^2 = \{w \in \mathbb{R}^3 : \langle w, N(p)\rangle = 0\}.$$

Step 2: T_pS and $T_{N(p)}S^2$ are the same 2-plane in $\mathbb{R}^3$. Because $N(p)$ is also the unit normal to S at p, the tangent plane to S at p is

$$T_pS = \{w \in \mathbb{R}^3 : \langle w, N(p)\rangle = 0\}.$$

But this is exactly the same description as the tangent plane to the unit sphere at $N(p)$:

$$T_{N(p)}S^2 = \{w \in \mathbb{R}^3 : \langle w, N(p)\rangle = 0\}.$$

Therefore, as subspaces of $\mathbb{R}^3$,

$$T_pS = T_{N(p)}S^2.$$

With this identification, we may regard

$$dN_p : T_pS \to T_pS.$$

Step 3: The shape operator. Using the above identification, one defines the *shape operator* (Weingarten map) by

$$S_p := -\,dN_p : T_pS \to T_pS.$$

Its eigenvectors are the *principal directions*, its eigenvalues are the *principal curvatures*, and one has

$$K(p) = \det(S_p), \qquad 2H(p) = \operatorname{tr}(S_p),$$

with the sign of H depending on the chosen orientation of N, while K does not.

Geometric analogy (circle) and what the Gauss map does

It is helpful to first look at the 1-dimensional analogue. Let $\gamma : I \to \mathbb{R}^2$ be a unit-speed plane curve, with unit tangent $T(s)$ and unit normal $n(s)$. Then $n(s) \in S^1$ for all s, so $n : I \to S^1$ is a "Gauss map" for the curve.

Circle picture. For each point $q \in S^1$, the tangent line of the circle at q is

$$T_q S^1 = \{w \in \mathbb{R}^2 : \langle w, q\rangle = 0\}.$$

Now note that the normal $n(s)$ is a unit vector, hence it is a point on the unit circle S^1. Differentiating $\langle n, n\rangle = 1$ gives $\langle n'(s), n(s)\rangle = 0$, so

$$n'(s) \in T_{n(s)} S^1.$$

Thus the derivative of the Gauss map of the curve sends the tangent direction $T(s)$ to a vector lying in the tangent line to the circle at the point $n(s)$.

Surface version. For an oriented surface $S \subset \mathbb{R}^3$, the Gauss map $N : S \to S^2$ assigns to each $p \in S$ the unit normal vector $N(p) \in S^2$. Exactly as above, differentiating $\langle N, N\rangle = 1$ implies

$$dN_p(v) \perp N(p) \quad \text{for all } v \in T_p S,$$

so

$$dN_p(v) \in T_{N(p)} S^2.$$

But $T_{N(p)} S^2$ is the plane orthogonal to $N(p)$, and the surface tangent plane $T_p S$ is also the plane orthogonal to the same vector $N(p)$. Hence

$$T_p S = T_{N(p)} S^2 \subset \mathbb{R}^3,$$

so we may view dN_p as a linear map from the tangent plane of the surface to itself. Up to a minus sign, this map is the shape operator $S_p = -dN_p$, which encodes the curvature.

Problem 12 — Gauss curvature via the map fN; curvature of an ellipsoid

Problem. Let $S \subset \mathbb{R}^3$ be an oriented smooth surface with Gauss map (unit normal) N. Let $V \subset S$ be open and let $f : V \to \mathbb{R}$ be smooth and nowhere zero. Let v_1, v_2 be smooth tangent vector fields on V such that at each point they are orthonormal and oriented, i.e. $v_1 \wedge v_2 = N$ (equivalently $v_1 \times v_2 = N$).

1. Prove that the Gaussian curvature K on V is

$$K \;=\; \frac{\big\langle D(fN)(v_1) \wedge D(fN)(v_2),\, fN\big\rangle}{f^3}.$$

2. Let f be the restriction to the ellipsoid

$$E : \qquad \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

of the function

$$f(x,y,z)=\sqrt{\frac{x^2}{a^4}+\frac{y^2}{b^4}+\frac{z^2}{c^4}}.$$

Show that the Gaussian curvature of E is

$$K=\frac{1}{a^2b^2c^2\,f^4}.$$

Theory. For an oriented orthonormal basis (v_1,v_2) of T_pS,

$$K(p)=\det(S_p)=\big\langle dN_p(v_1)\times dN_p(v_2),\,N(p)\big\rangle,$$

because $S=-dN$ and $\det(-S)=\det(S)$. Also $\langle N\times X,\,N\rangle=0$ for all X.

Solution.

(i) Set $F=fN:V\to\mathbb{R}^3$. For any $v\in T_pS$,

$$dF_p(v)=v(f)\,N(p)+f(p)\,dN_p(v).$$

Hence

$$\begin{aligned}dF(v_1)\times dF(v_2)&=\big(v_1(f)N+f\,dN(v_1)\big)\times\big(v_2(f)N+f\,dN(v_2)\big)\\&=f^2\,dN(v_1)\times dN(v_2)+f\,v_1(f)\,N\times dN(v_2)+f\,v_2(f)\,dN(v_1)\times N.\end{aligned}$$

Taking the dot product with fN kills the mixed terms since $\langle N\times dN(v_2),N\rangle=0$ and $\langle dN(v_1)\times N,N\rangle=0$. Thus

$$\big\langle dF(v_1)\times dF(v_2),\,fN\big\rangle=f^3\,\big\langle dN(v_1)\times dN(v_2),\,N\big\rangle.$$

But for an oriented orthonormal pair (v_1,v_2), the Gauss curvature is

$$K=\big\langle dN(v_1)\times dN(v_2),\,N\big\rangle.$$

Therefore

$$K=\frac{\big\langle dF(v_1)\times dF(v_2),\,fN\big\rangle}{f^3}=\frac{\big\langle D(fN)(v_1)\wedge D(fN)(v_2),\,fN\big\rangle}{f^3}.$$

(ii) Let $\Phi(x,y,z)=\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}$, so $E=\{\Phi=1\}$ and

$$\nabla\Phi=\left(\frac{2x}{a^2},\frac{2y}{b^2},\frac{2z}{c^2}\right),\qquad\|\nabla\Phi\|=2\sqrt{\frac{x^2}{a^4}+\frac{y^2}{b^4}+\frac{z^2}{c^4}}=2f.$$

Hence the unit normal is

$$N=\frac{\nabla\Phi}{\|\nabla\Phi\|}=\frac{1}{f}\left(\frac{x}{a^2},\frac{y}{b^2},\frac{z}{c^2}\right),\quad\text{so}\quad fN=\left(\frac{x}{a^2},\frac{y}{b^2},\frac{z}{c^2}\right).$$

Write $A=\operatorname{diag}\big(\frac{1}{a^2},\frac{1}{b^2},\frac{1}{c^2}\big)$. Then $fN=Ax$ and thus

$$D(fN)(v)=Av.$$

Applying (i),

$$K=\frac{\langle (Av_1)\times(Av_2),\,fN\rangle}{f^3}.$$

Use the identity $(Au) \times (Av) = \det(A)\, A^{-T}(u \times v)$. Since A is diagonal, $A^{-T} = A^{-1}$, and $v_1 \times v_2 = N$, we obtain

$$(Av_1) \times (Av_2) = \det(A)\, A^{-1}N.$$

Therefore

$$\langle (Av_1) \times (Av_2),\, fN \rangle = \det(A)\, \langle A^{-1}N,\, fN \rangle.$$

But $N = \frac{1}{f}Ax$, hence $A^{-1}N = \frac{1}{f}x$, and $fN = Ax$, so

$$\langle A^{-1}N,\, fN \rangle = \left\langle \frac{1}{f}x,\, Ax \right\rangle = \frac{1}{f}\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right) = \frac{1}{f}$$

on E. Also $\det(A) = \frac{1}{a^2b^2c^2}$. Hence

$$\langle (Av_1) \times (Av_2),\, fN \rangle = \frac{1}{a^2b^2c^2\, f}, \qquad K = \frac{1}{a^2b^2c^2\, f^4}.$$

Example (sphere check). If $a = b = c = R$, then $f = 1/R$ on E, so $K = \frac{1}{R^6(1/R^4)} = \frac{1}{R^2}$.

Problem 12 — Relation to the "area" definition of Gaussian curvature; meaning of N and fN

What is N? Let $S \subset \mathbb{R}^3$ be an oriented smooth surface. At each point $p \in S$ there are exactly two unit normals; an *orientation* chooses one of them smoothly. The *Gauss map* (or *unit normal field*) is the smooth map

$$N : S \to S^2, \qquad N(p) = \text{the chosen unit normal at } p.$$

Thus $N(p)$ is a unit vector in $\mathbb{R}^3$ perpendicular to the tangent plane T_pS. Equivalently, N is a vector field along S with values on the unit sphere S^2.

How does N give Gaussian curvature via area? For a small patch $B \subset S$ around p, the Gauss map sends it to a patch $N(B) \subset S^2$. A classical interpretation of Gaussian curvature is that it is the infinitesimal *area distortion* of the Gauss map:

$$K(p) = \lim_{B \to p} \frac{\text{Area}(N(B))}{\text{Area}(B)} \quad \text{(with sign depending on orientation).}$$

In differential form: if (v_1, v_2) is an oriented orthonormal basis of T_pS, then

$$K(p) = \big\langle dN_p(v_1) \times dN_p(v_2),\, N(p) \big\rangle.$$

This is just the Jacobian determinant of dN_p written as a triple product.

What is fN? Let $f : V \to \mathbb{R}$ be a smooth function on an open set $V \subset S$ such that $f \neq 0$ everywhere. Then

$$F := fN : V \to \mathbb{R}^3, \qquad F(p) = f(p)\, N(p),$$

is the normal vector field N *scaled* by the factor f at each point. Geometrically, $fN(p)$ is a normal vector at p whose length is $|f(p)|$ (instead of length 1). Unlike N, the map fN does *not* take values in S^2; it takes values in $\mathbb{R}^3 \setminus \{0\}$ (on rays through the origin).

Why does the formula in Problem 12 give the same curvature as the area definition? Set $F = fN$. For any tangent vector $v \in T_pS$,

$$dF_p(v) = v(f)\, N(p) + f(p)\, dN_p(v).$$

If (v_1, v_2) is oriented orthonormal, expand the cross product and dot with fN:

$$\big\langle dF(v_1) \times dF(v_2),\, fN \big\rangle = \big\langle (v_1(f)N + f\, dN(v_1)) \times (v_2(f)N + f\, dN(v_2)),\, fN \big\rangle.$$

All mixed terms vanish after dotting with N because $\langle N \times X,\, N\rangle = 0$ for every X. Hence only the $f^2(dN(v_1) \times dN(v_2))$ term survives, giving

$$\big\langle dF(v_1) \times dF(v_2),\, fN \big\rangle = f^3 \big\langle dN(v_1) \times dN(v_2),\, N \big\rangle = f^3\, K.$$

Therefore

$$K = \frac{\big\langle D(fN)(v_1) \wedge D(fN)(v_2),\, fN \big\rangle}{f^3},$$

which is exactly the same Gaussian curvature K as the "area ratio" definition (it is still $\det(dN_p)$); the map fN is merely a convenient re-packaging that cancels out to K after dividing by f^3.

Why is fN useful for the ellipsoid? For the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, one chooses

$$f(x, y, z) = \sqrt{\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4}}$$

so that

$$fN = \left(\frac{x}{a^2}, \frac{y}{b^2}, \frac{z}{c^2}\right),$$

which is the restriction of a *linear map* $\mathbb{R}^3 \to \mathbb{R}^3$. Then $D(fN)$ is constant, and the curvature computation becomes algebraic.

Problem 1 — Plane geodesics are principal directions

Problem. Let $\alpha : I \to S$ be a geodesic on an oriented surface $S \subset \mathbb{R}^3$ with Gauss map (unit normal) N. Assume that α is a plane curve and that $\ddot{\alpha}(t_0) \neq 0$ for some $t_0 \in I$. Show that $\dot{\alpha}(t_0)$ is an eigenvector of the differential of the Gauss map $dN_{\alpha(t_0)}$.

Solution. Work locally with an arc-length parameter s (reparametrize near t_0 if needed), so that $T(s) = \alpha'(s)$ satisfies $\|T(s)\| = 1$. Then

$$T'(s) = \alpha''(s) = \kappa(s)\, n(s),$$

where κ is the curvature and n is the principal normal of α as a space curve. Since $\ddot{\alpha}(t_0) \neq 0$, we have $\kappa(s_0) \neq 0$ at the corresponding s_0, hence $n(s_0)$ is defined.

Because α is a geodesic, its geodesic curvature is zero, so the acceleration is purely normal to the surface:

$$T'(s) \ \parallel \ N(\alpha(s)).$$

Therefore, wherever $\kappa(s) \neq 0$,

$$n(s) = \pm N(\alpha(s)).$$

Since α is a plane curve, its binormal

$$b(s) = T(s) \times n(s)$$

is constant (it equals the fixed unit normal to the plane), hence $b'(s) = 0$. On any interval where $\kappa \neq 0$ we have $n = \pm N$, so

$$b(s) = T(s) \times n(s) = \pm\, T(s) \times N(\alpha(s)).$$

Differentiate:

$$0 = b'(s) = \pm\big(T'(s) \times N + T \times N'(s)\big).$$

But $T'(s) \parallel N$, so $T'(s) \times N = 0$, and thus

$$T(s) \times N'(s) = 0,$$

which implies $N'(s)$ is parallel to $T(s)$. By the chain rule,

$$N'(s) = dN_{\alpha(s)}(T(s)).$$

Hence $dN_{\alpha(s)}(T(s))$ is a scalar multiple of $T(s)$, i.e.

$$dN_{\alpha(s)}(T(s)) = \lambda(s)\, T(s),$$

so $T(s) = \dot\alpha(s)$ is an eigenvector of $dN_{\alpha(s)}$ (equivalently, a principal direction). In particular, $\dot\alpha(t_0)$ is an eigenvector of $dN_{\alpha(t_0)}$.

Problem 2: Planar geodesics imply plane or sphere

Statement. Show that if all geodesics of a connected surface $S \subset \mathbb{R}^3$ are plane curves, then S is contained in a plane or a sphere.

Solution. Let $\alpha(s)$ be a unit-speed geodesic on S, with unit tangent $T = \alpha'(s)$ and unit normal field n of the surface along α.

Since α is a geodesic, its acceleration is purely normal to the surface:

$$\alpha''(s) = \kappa_n(s)\, n(s).$$

Hence the principal normal N of the space curve α equals $\pm n$, and the binormal is

$$B = T \times N = \pm\, T \times n.$$

A unit-speed space curve is planar iff its torsion vanishes, equivalently iff B is constant. Thus, for a planar geodesic we have $B' \equiv 0$. Compute

$$B' = (T \times n)' = T' \times n + T \times n'.$$

Using $T' = \alpha'' = \kappa_n n$, we get $T' \times n = 0$, so

$$B' = T \times n'.$$

By the Weingarten equation, $n' = dn(T) = -A(T)$ where A is the shape operator. Therefore

$$B' = -\,T \times A(T).$$

If $B' \equiv 0$ then $T \times A(T) = 0$, hence $A(T)$ is parallel to T; i.e. the tangent direction of α is a principal direction.

Now fix $p \in S$ and any $v \in T_pS$. There exists a (local) geodesic α with $\alpha(0) = p$ and $\alpha'(0) = v$. Applying the previous conclusion at $s = 0$ yields $A_p(v) \parallel v$ for all $v \in T_pS$. Hence

$$A_p = \lambda(p)\, I \quad \text{for every } p \in S,$$

so S is totally umbilic.

For a totally umbilic surface, the Codazzi equations imply $\nabla\lambda = 0$, hence λ is constant on the connected surface S. If $\lambda = 0$, then $A = 0$ and the normal is constant, so S is contained in a plane. If $\lambda \neq 0$, then $dn = -\lambda\, dX$ and therefore

$$d(n + \lambda X) = dn + \lambda\, dX = 0,$$

so $n + \lambda X = c$ is constant. Hence

$$X - \frac{c}{\lambda} = -\frac{1}{\lambda} n,$$

and taking norms gives $\left\| X - \frac{c}{\lambda} \right\| = \frac{1}{|\lambda|}$, i.e. S is contained in a sphere. □

Problem 3: Isometries preserve parallel transport and geodesics

Statement. Let $f : S_1 \to S_2$ be an isometry between two surfaces.

1. Let $\alpha : I \to S_1$ be a curve and V a vector field along α. Let $\gamma := f \circ \alpha$, and define a vector field along γ by
$$W(t) := Df_{\alpha(t)}\big(V(t)\big).$$
Show that
$$\frac{DW}{dt} = Df_{\alpha(t)}\left(\frac{DV}{dt}\right),$$
and hence that V parallel along α implies that W is parallel along γ.

2. Deduce that f maps geodesics to geodesics.

1. **Statement.** Let $\alpha : I \to S_1$ be a curve and V a vector field along α. Let $\gamma = f \circ \alpha$ and define $W(t) = Df_{\alpha(t)}(V(t))$, a vector field along γ. Show that
$$\frac{DW}{dt} = Df_{\alpha(t)}\left(\frac{DV}{dt}\right),$$
and deduce that V parallel along α implies W parallel along γ.

   Solution. Let $\nabla^{(1)}$ and $\nabla^{(2)}$ be the Levi–Civita connections on S_1 and S_2. Define a connection $\widetilde{\nabla}$ on S_1 by
$$\widetilde{\nabla}_X Y := Df^{-1}\Big(\nabla^{(2)}_{Df(X)}(Df(Y))\Big).$$
Because f is an isometry, $\widetilde{\nabla}$ is metric-compatible; and since $\nabla^{(2)}$ is torsion-free and Df preserves Lie brackets, $\widetilde{\nabla}$ is torsion-free. By uniqueness of the Levi–Civita connection, $\widetilde{\nabla} = \nabla^{(1)}$, hence for all vector fields X, Y on S_1,
$$\nabla^{(2)}_{Df(X)}(Df(Y)) = Df\big(\nabla^{(1)}_X Y\big).$$

Apply this along the curve α with $X = \alpha'(t)$ and $Y = V(t)$:

$$\nabla^{(2)}_{\gamma'(t)} W(t) = Df_{\alpha(t)}\big(\nabla^{(1)}_{\alpha'(t)} V(t)\big).$$

By definition, $\frac{DV}{dt} = \nabla^{(1)}_{\alpha'} V$ and $\frac{DW}{dt} = \nabla^{(2)}_{\gamma'} W$, so

$$\frac{DW}{dt} = Df_{\alpha(t)}\left(\frac{DV}{dt}\right).$$

In particular, if $\frac{DV}{dt} = 0$ then $\frac{DW}{dt} = 0$.

2. **Statement.** Deduce that f maps geodesics to geodesics.

   Solution. If α is a geodesic on S_1, then its tangent field $T = \alpha'$ is parallel:

$$\nabla^{(1)}_{\alpha'} \alpha' = 0.$$

   Let $\gamma = f \circ \alpha$. Then $\gamma' = Df(\alpha')$. By part (i), $\nabla^{(2)}_{\gamma'} \gamma' = 0$, hence γ is a geodesic on S_2. □

Theory (Problem 3)

1. **Isometry and pushforward.** An isometry $f : S_1 \to S_2$ preserves the first fundamental form, i.e. for every $p \in S_1$ and all $u, v \in T_pS_1$,

$$\langle Df_p(u), Df_p(v)\rangle_{S_2} = \langle u, v\rangle_{S_1}.$$

   Hence $Df_p : T_pS_1 \to T_{f(p)}S_2$ is an inner-product preserving linear map.

2. **Isometries preserve the Levi–Civita connection.** Let $\nabla^{(1)}$ and $\nabla^{(2)}$ be the Levi–Civita connections on S_1 and S_2. Define a connection $\widetilde{\nabla}$ on S_1 by

$$\widetilde{\nabla}_X Y := Df^{-1}\Big(\nabla^{(2)}_{Df(X)}(Df(Y))\Big).$$

   Because f is an isometry, $\widetilde{\nabla}$ is metric-compatible; and since $\nabla^{(2)}$ is torsion-free and Df preserves Lie brackets, $\widetilde{\nabla}$ is torsion-free. By uniqueness of the Levi–Civita connection, $\widetilde{\nabla} = \nabla^{(1)}$. Equivalently, for all vector fields X, Y on S_1,

$$\nabla^{(2)}_{Df(X)}(Df(Y)) \;=\; Df\big(\nabla^{(1)}_X Y\big).$$

3. **Parallel vector fields are mapped to parallel vector fields.** Let $\alpha : I \to S_1$ be a curve and V a vector field along α. Put $\gamma = f \circ \alpha$ and $W(t) := Df_{\alpha(t)}(V(t))$. Using the identity in (2) with $X = \alpha'(t)$ and $Y = V(t)$, we obtain

$$\frac{DW}{dt} = \nabla^{(2)}_{\gamma'(t)} W(t) = Df_{\alpha(t)}\Big(\nabla^{(1)}_{\alpha'(t)} V(t)\Big) = Df_{\alpha(t)}\left(\frac{DV}{dt}\right).$$

   In particular, if V is parallel along α (i.e. $\frac{DV}{dt} = 0$), then W is parallel along γ (i.e. $\frac{DW}{dt} = 0$).

4. **Geodesics are mapped to geodesics.** A curve α on S_1 is a geodesic iff its tangent field is parallel:
$$\nabla^{(1)}_{\alpha'}\alpha' = 0.$$
Let $\gamma = f \circ \alpha$. Since $\gamma' = Df(\alpha')$, applying (3) to $V = \alpha'$ yields
$$\nabla^{(2)}_{\gamma'}\gamma' = 0,$$
so γ is a geodesic on S_2.

$$\begin{tikzcd}
T_{\alpha(t)}S_1 \arrow[r, "Df_{\alpha(t)}"] \arrow[d, "\nabla^{(1)}_{\alpha'(t)}"'] & T_{\gamma(t)}S_2 \arrow[d, "\nabla^{(2)}_{\gamma'(t)}"] \\
T_{\alpha(t)}S_1 \arrow[r, "Df_{\alpha(t)}"'] & T_{\gamma(t)}S_2
\end{tikzcd}$$

$$\begin{tikzcd}
V(t) \in T_{\alpha(t)}S_1 \arrow[r, "Df_{\alpha(t)}"] \arrow[d, "\frac{D}{dt}"'] & W(t) \in T_{\gamma(t)}S_2 \arrow[d, "\frac{D}{dt}"] \\
\frac{DV}{dt}(t) \arrow[r, "Df_{\alpha(t)}"'] & \frac{DW}{dt}(t)
\end{tikzcd}$$
where $W(t) = Df_{\alpha(t)}(V(t)),\ \gamma = f \circ \alpha.$

$$\nabla^{(1)}_{\alpha'}\alpha' = 0 \xrightarrow{Df \text{ and (i)}} \nabla^{(2)}_{\gamma'}\gamma' = 0 \qquad (\gamma = f \circ \alpha)$$

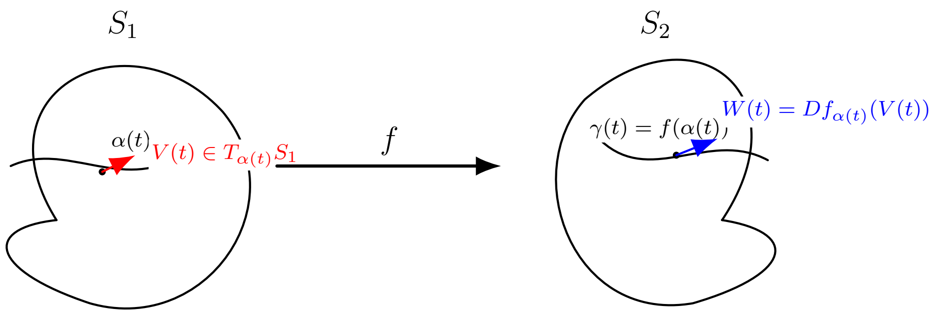


Figure 4: An isometry f pushes forward a vector field V along α to the vector field $W(t) = Df_{\alpha(t)}(V(t))$ along $\gamma = f \circ \alpha$.

Levi–Civita connection

Let (M, g) be a Riemannian manifold (in particular, a regular surface with the first fundamental form). A *connection* on TM is a map
$$\nabla : \Gamma(TM) \times \Gamma(TM) \to \Gamma(TM), \qquad (X, Y) \mapsto \nabla_X Y,$$
called the *covariant derivative*, such that for all smooth functions f and vector fields X, Y, Z:

1. $\nabla_{fX+Y} Z = f\nabla_X Z + \nabla_Y Z$,
2. $\nabla_X (Y + Z) = \nabla_X Y + \nabla_X Z$,
3. $\nabla_X (fY) = X(f)\, Y + f\nabla_X Y$.

There exists a unique connection ∇ satisfying:

1. **Metric compatibility** ($\nabla g = 0$):
$$X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle,$$
2. **Torsion-free**:
$$\nabla_X Y - \nabla_Y X = [X, Y].$$

This unique connection is the *Levi–Civita connection* of (M, g).

If $S \subset \mathbb{R}^3$ is a surface with unit normal n and $\bar{\nabla}$ denotes the usual derivative in $\mathbb{R}^3$, then for tangent vector fields X, Y on S one has the decomposition

$$\bar{\nabla}_X Y = \nabla_X Y + \mathrm{II}(X, Y)\, n, \qquad \mathrm{II}(X, Y) = \langle \bar{\nabla}_X Y, n\rangle,$$

so $\nabla_X Y$ is the tangential part of the ambient derivative.

For a curve $\alpha : I \to M$ and a vector field V along α, the covariant derivative along α is

$$\frac{DV}{dt} := \nabla_{\alpha'(t)} V.$$

A field V is *parallel* along α if $\frac{DV}{dt} = 0$, and α is a *geodesic* if

$$\frac{D\alpha'}{dt} = \nabla_{\alpha'}\alpha' = 0.$$

In local coordinates (u^1, u^2) with $\partial_i = \frac{\partial}{\partial u^i}$ one writes

$$\nabla_{\partial_i}\partial_j = \Gamma^k_{ij}\, \partial_k, \qquad \nabla_{\partial_i}(Y^j \partial_j) = \left(\frac{\partial Y^k}{\partial u^i} + \Gamma^k_{ij} Y^j \right) \partial_k.$$

Explanation of the notation

- **The tangent bundle and vector fields.** For a smooth manifold (or surface) M, the *tangent bundle* TM is the union of all tangent spaces T_pM over $p \in M$. The notation $\Gamma(TM)$ denotes the set of smooth sections of TM, i.e. the set of smooth *vector fields* on M:
$$X \in \Gamma(TM) \iff X(p) \in T_pM \text{ for all } p \in M, \text{ smoothly in } p.$$
- **A connection (covariant derivative).** A (linear) connection on TM is a map
$$\nabla : \Gamma(TM) \times \Gamma(TM) \to \Gamma(TM), \qquad (X, Y) \mapsto \nabla_X Y,$$
which should be thought of as "differentiate the vector field Y in the direction of the vector field X". It satisfies the usual linearity and Leibniz rules:
$$\nabla_{fX+gX'} Y = f\nabla_X Y + g\nabla_{X'} Y, \qquad \nabla_X (Y + Y') = \nabla_X Y + \nabla_X Y', \qquad \nabla_X (fY) = X(f)\, Y + f\nabla_X Y.$$

- **The curve $\gamma = f \circ \alpha$.** If $\alpha : I \to S_1$ is a curve and $f : S_1 \to S_2$ is a smooth map, then the *image curve*

$$\gamma := f \circ \alpha : I \to S_2, \qquad \gamma(t) = f(\alpha(t)).$$

  By the chain rule, the tangent vectors satisfy

$$\gamma'(t) = Df_{\alpha(t)}\big(\alpha'(t)\big),$$

  where $Df_{\alpha(t)} : T_{\alpha(t)}S_1 \to T_{\gamma(t)}S_2$ is the differential (pushforward) of f at $\alpha(t)$.

A (smooth) vector field on M means a smooth *tangent* vector field, i.e. a section of the tangent bundle:

$$X \in \Gamma(TM) \quad \Longleftrightarrow \quad X : M \to TM, \;\; X(p) \in T_pM \text{ for all } p \in M.$$

If $M \subset \mathbb{R}^3$ is embedded, an ambient vector field $Y : M \to \mathbb{R}^3$ need not be tangent; it splits as $Y = Y^\top + Y^\perp$ into tangent and normal components.

Example (sphere). Let $M = S^2 \subset \mathbb{R}^3$ be the unit sphere. Its unit normal at $p \in S^2$ is $n(p) = p$. Consider the ambient vector field $Y(p) = e_3 = (0, 0, 1)$. In general Y is not tangent to S^2. The normal component is the projection onto $n(p)$:

$$Y^\perp(p) = \langle Y(p), n(p)\rangle n(p) = \langle e_3, p\rangle\, p = p_3\, p,$$

and therefore the tangential component is

$$Y^\top(p) = Y(p) - Y^\perp(p) = e_3 - p_3\, p.$$

One checks tangency by $\langle Y^\top(p), p\rangle = 0$, hence $Y^\top(p) \in T_pS^2$.

Example (plane). Let $M = \{(x, y, 0)\} \subset \mathbb{R}^3$ with unit normal $n = e_3$. For the ambient field $Y \equiv e_3$ we get

$$Y^\perp = \langle e_3, n\rangle n = e_3, \qquad Y^\top = Y - Y^\perp = 0,$$

so Y is purely normal to the plane.

Example (graph surface). Let M be the graph $X(x, y) = (x, y, f(x, y))$. Then $X_x = (1, 0, f_x)$, $X_y = (0, 1, f_y)$ and a normal is $N = X_x \times X_y = (-f_x, -f_y, 1)$. For an ambient constant field $Y \equiv e_1 = (1, 0, 0)$ one has the decomposition

$$Y^\perp = \langle Y, n\rangle n, \qquad Y^\top = Y - \langle Y, n\rangle n, \quad \text{where } n = \frac{N}{\|N\|}.$$

Unless $f_x \equiv 0$, the field Y is not tangent to M.

A (tangent) vector field on M is a smooth section of the tangent bundle because it assigns to each point $p \in M$ a vector based at p lying in the tangent space at p:

$$X \in \Gamma(TM) \quad \Longleftrightarrow \quad X : M \to TM, \;\; X(p) \in T_pM \text{ for all } p, \text{ and } \pi \circ X = \mathrm{id}_M,$$

where $\pi : TM \to M$ is the bundle projection. For a surface, T_pM is the tangent plane at p.

$\Gamma(TM)$ denotes the set of smooth tangent vector fields on M, i.e. maps $X : M \to TM$ with $X(p) \in T_pM$ for every $p \in M$.

A connection is a map

$$\nabla : \Gamma(TM) \times \Gamma(TM) \to \Gamma(TM), \qquad (X, Y) \mapsto \nabla_X Y,$$

so $\nabla_X Y$ is again a tangent vector field. In particular, for each $p \in M$,

$$(\nabla_X Y)(p) \in T_pM.$$

Heuristically, $(\nabla_X Y_b t)(p)$ measures the variation of Y at p in the direction $X(p)$. Along a curve $\alpha(t)$ and a field $V(t) \in T_{\alpha(t)}M$, one writes

$$\frac{DV}{dt} := \nabla_{\alpha'(t)} V \in T_{\alpha(t)}M.$$

The covariant derivative $\nabla_X Y$ is the "directional derivative" of the vector field Y in the direction X, defined intrinsically on M. It is a vector field, so for each $p \in M$ one has

$$(\nabla_X Y)(p) \in T_pM.$$

Geometrically, $(\nabla_X Y)(p)$ measures how Y changes when one moves away from p in the direction $X(p)$, using the connection to compare vectors in different tangent spaces.

Equivalently, if $c(t)$ is any curve with $c(0) = p$ and $c'(0) = X(p)$ and $P_t : T_pM \to T_{c(t)}M$ denotes parallel transport along c, then

$$(\nabla_X Y)(p) = \left.\frac{d}{dt}\right|_{t=0} \Big(P_t^{-1}(Y(c(t)))\Big).$$

Along a curve $\alpha(t)$ and a field $V(t) \in T_{\alpha(t)}M$, one writes

$$\frac{DV}{dt} := \nabla_{\alpha'(t)} V,$$

so $DV/dt = 0$ means that V is parallel along α, and $\nabla_{\alpha'}\alpha' = 0$ means that α is a geodesic.

Example on S^2: computing $\nabla_{\alpha'} V$ by projection

If $S \subset \mathbb{R}^3$ is a surface with unit normal n, then along a curve $\alpha(t) \subset S$ and a tangent field $V(t) \in T_{\alpha(t)}S$ one can compute the Levi–Civita covariant derivative by

$$\frac{DV}{dt} = \left(\frac{dV}{dt}\right)^{\top} = \frac{dV}{dt} - \left\langle \frac{dV}{dt},\, n(\alpha(t)) \right\rangle n(\alpha(t)).$$

For the unit sphere S^2, one has $n(p) = p$.

(A) Great circle (equator) is a geodesic. Let $\alpha(s) = (\cos s, \sin s, 0)$ (arc-length parametrization). Then

$$T(s) = \alpha'(s) = (-\sin s, \cos s, 0), \qquad \frac{dT}{ds} = (-\cos s, -\sin s, 0).$$

Since $n(\alpha(s)) = \alpha(s)$,

$$\left\langle \frac{dT}{ds},\, \alpha(s) \right\rangle = (-\cos s, -\sin s, 0) \cdot (\cos s, \sin s, 0) = -1.$$

Hence

$$\frac{DT}{ds} = \frac{dT}{ds} - \left\langle \frac{dT}{ds}, \alpha \right\rangle \alpha = (-\cos s, -\sin s, 0) - (-1)\,(\cos s, \sin s, 0) = 0,$$

so $\nabla_T T = 0$ and the equator is a geodesic.

(B) Latitude is not a geodesic (except the equator). Fix $\theta \in (0, \pi/2)$ and consider

$$\alpha(\varphi) = \big(\cos\theta\cos\varphi,\ \cos\theta\sin\varphi,\ \sin\theta\big).$$

Then $|\alpha'(\varphi)| = \cos\theta$, and the unit tangent is

$$T(\varphi) = \frac{\alpha'(\varphi)}{|\alpha'(\varphi)|} = (-\sin\varphi, \cos\varphi, 0).$$

Differentiate:

$$\frac{dT}{d\varphi} = (-\cos\varphi, -\sin\varphi, 0), \qquad \left\langle \frac{dT}{d\varphi},\, \alpha(\varphi) \right\rangle = -\cos\theta.$$

Therefore the tangential projection is

$$\left(\frac{dT}{d\varphi}\right)^{\top} = \frac{dT}{d\varphi} - \left\langle \frac{dT}{d\varphi}, \alpha \right\rangle \alpha = (-\cos\varphi, -\sin\varphi, 0) + \cos\theta\, \alpha(\varphi) \neq 0 \quad (\theta \neq 0).$$

Since $ds = \cos\theta\, d\varphi$, we have

$$\frac{DT}{ds} = \frac{1}{\cos\theta}\left(\frac{dT}{d\varphi}\right)^{\top} \neq 0 \quad (\theta \neq 0),$$

so a latitude circle is not a geodesic, except when $\theta = 0$ (the equator).

How to understand a connection (Earth / sphere intuition)

Think of $M = S^2$ (Earth idealized as a sphere). A wind (velocity) field is a *tangent* vector field

$$Y \in \Gamma(TM), \qquad Y(p) \in T_pS^2 \text{ for each } p \in S^2,$$

so at every point you have an arrow lying in the local tangent plane.

Why we need a connection. If p and q are two nearby points, then T_pS^2 and T_qS^2 are different tangent planes. Hence one cannot directly form $Y(q) - Y(p)$, because these vectors live in different vector spaces.

Role of the connection. A connection ∇ provides a rule to compare vectors in different tangent planes and therefore defines a directional derivative of vector fields. Given a tangent direction $X(p) \in T_pS^2$, the covariant derivative

$$(\nabla_X Y)(p) \in T_pS^2$$

measures how Y changes at p when one moves in the direction $X(p)$, after correcting for the curvature of the surface.

Parallel transport interpretation. Let $c(t)$ be a curve on S^2 with $c(0) = p$ and $c'(0) = X(p)$, and let $P_t : T_pS^2 \to T_{c(t)}S^2$ denote parallel transport along c (for the Levi–Civita connection). Then

$$(\nabla_X Y)(p) = \frac{d}{dt}\bigg|_{t=0} \Big(P_t^{-1}(Y(c(t)))\Big).$$

Here $P_t^{-1}(Y(c(t)))$ is the vector $Y(c(t))$ moved back to the original tangent space T_pS^2, so the usual difference quotient makes sense.

Geodesics. For a curve $c(t)$, its velocity $T = c'(t)$ is tangent. The curve is a geodesic iff

$$\nabla_T T = 0,$$

meaning its tangent vector is parallel along the curve ("straightest possible motion on the surface").

How to understand a connection (Earth / sphere intuition)

Think of $M = S^2$ (Earth idealized as a sphere). A wind (velocity) field is a *tangent* vector field

$$Y \in \Gamma(TM), \qquad Y(p) \in T_pS^2 \text{ for each } p \in S^2,$$

so at every point you have an arrow lying in the local tangent plane.

Why we need a connection. If p and q are two nearby points, then T_pS^2 and T_qS^2 are different tangent planes. Hence one cannot directly form $Y(q) - Y(p)$, because these vectors live in different vector spaces.

Role of the connection. A connection ∇ provides a rule to compare vectors in different tangent planes and therefore defines a directional derivative of vector fields. Given a tangent direction $X(p) \in T_pS^2$, the covariant derivative

$$(\nabla_X Y)(p) \in T_pS^2$$

measures how Y changes at p when one moves in the direction $X(p)$, after correcting for the curvature of the surface.

Parallel transport interpretation. Let $c(t)$ be a curve on S^2 with $c(0) = p$ and $c'(0) = X(p)$, and let $P_t : T_pS^2 \to T_{c(t)}S^2$ denote parallel transport along c (for the Levi–Civita connection). Then

$$(\nabla_X Y)(p) = \frac{d}{dt}\bigg|_{t=0} \Big(P_t^{-1}(Y(c(t)))\Big).$$

Here $P_t^{-1}(Y(c(t)))$ is the vector $Y(c(t))$ moved back to the original tangent space T_pS^2, so the usual difference quotient makes sense.

Geodesics. For a curve $c(t)$, its velocity $T = c'(t)$ is tangent. The curve is a geodesic iff

$$\nabla_T T = 0,$$

meaning its tangent vector is parallel along the curve ("straightest possible motion on the surface").

What the connection gives you

A connection ∇ provides a consistent way to:

1. **compare vectors in different tangent planes**, and therefore
2. **define a derivative** of a vector field.

"Derivative in a direction" (the meaning of $\nabla_X Y$)

Pick:

- a direction of motion $X(p) \in T_pS^2$ (for example: your walking direction),
- and a tangent vector field Y (for example: wind velocity).

Then $(\nabla_X Y)(p) \in T_pS^2$ means:

Start at p, move a tiny bit in the direction $X(p)$. Bring the vector Y at the new point back to the original tangent plane (using the connection's *"keep direction as constant as possible"* rule). The rate of change of the difference is $(\nabla_X Y)(p)$.

Parallel transport formula. Let $c(t)$ be a curve with $c(0) = p$ and $c'(0) = X(p)$, and let $P_t : T_pS^2 \to T_{c(t)}S^2$ denote parallel transport along c. Then

$$(\nabla_X Y)(p) = \frac{d}{dt}\bigg|_{t=0} \Big(P_t^{-1}(Y(c(t)))\Big).$$

Chart 1: the spherical-triangle loop (holonomy picture)

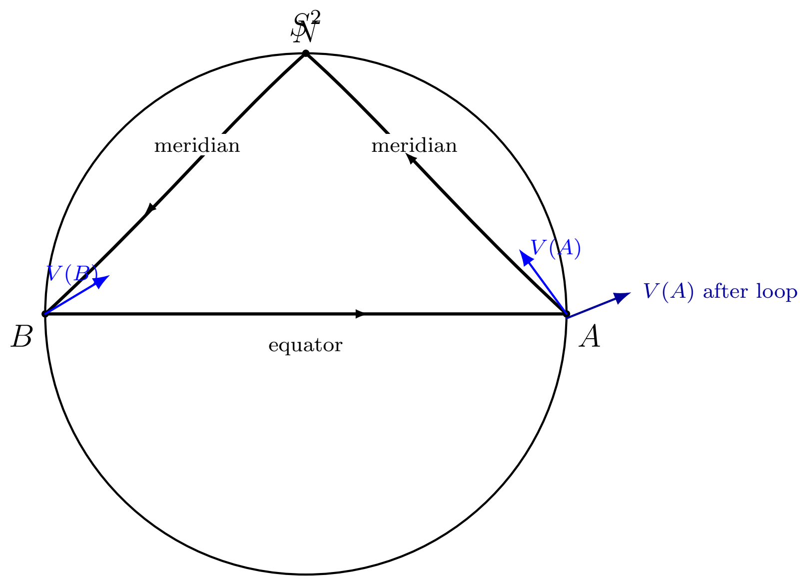


Figure 5: Schematic spherical triangle loop $A \to N \to B \to A$. Parallel transport ($\nabla_{c'}V = 0$ along each edge) can rotate the vector after returning to A (*holonomy*), reflecting curvature.

Chart 2: covariant derivative via parallel transport (diagram)

$$\begin{tikzcd}
T_pM \arrow[r, "P_t"] & T_{c(t)}M \\
Y(p) \arrow[r, mapsto] & Y(c(t)) \arrow[d, "P_t^{-1}"] \\
\text{compare in } T_pM & P_t^{-1}\big(Y(c(t))\big) \in T_pM
\end{tikzcd}$$

$$(\nabla_X Y)(p) = \frac{d}{dt}\bigg|_{t=0} \Big(P_t^{-1}(Y(c(t)))\Big), \qquad c(0) = p,\ c'(0) = X(p).$$

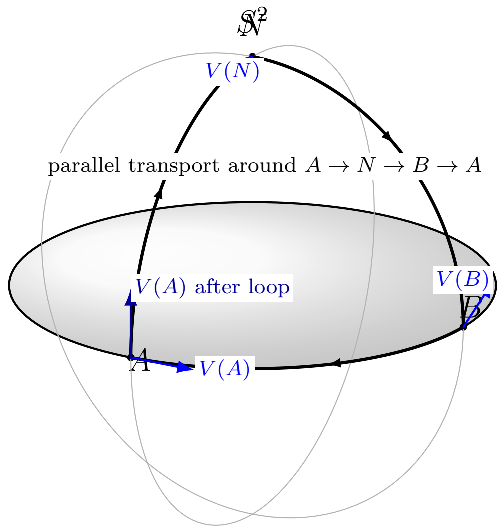


Figure 6: A 3D-looking schematic of a spherical triangle loop. A tangent vector field transported with $\nabla_{c'(t)}V = 0$ along the loop can return rotated at A (holonomy), reflecting curvature.

Parallel transport and directional derivative of a tangent field

On a manifold (in particular on a surface) tangent vectors at different points live in different vector spaces. Thus, for a curve $c(t)$ with $c(0) = p$, the expression $Y(c(t)) - Y(p)$ does not make sense intrinsically because $Y(c(t)) \in T_{c(t)}M$ while $Y(p) \in T_pM$.

Let $P_t : T_pM \to T_{c(t)}M$ denote the (Levi–Civita) parallel transport along c from 0 to t. Then $P_t^{-1} : T_{c(t)}M \to T_pM$ brings vectors back to T_pM, so that subtraction and differentiation become meaningful.

Definition (covariant derivative via parallel transport). Let $X(p) = c'(0) \in T_pM$ and let Y be a tangent vector field along c. The covariant derivative of Y in the direction X at p is

$$(\nabla_X Y)(p) := \frac{d}{dt}\bigg|_{t=0} \Big(P_t^{-1}(Y(c(t)))\Big).$$

Equivalently: transport $Y(c(t))$ back to T_pM without twisting (parallel transport), then differentiate in the fixed vector space T_pM.

Directional derivative: scalar vs. vector. For a scalar field $f : M \to \mathbb{R}$, the directional derivative is

$$X(f)(p) = \frac{d}{dt}\bigg|_{t=0} f(c(t)).$$

For a tangent vector field Y, the intrinsic directional derivative is $\nabla_X Y$, because one must compare vectors lying in different tangent planes.

Example (Earth as a sphere). Let $M = S^2$ model the Earth. Let X be a velocity field of motion on S^2 and let Y be a wind field, so $X(p), Y(p) \in T_pS^2$ for each $p \in S^2$. Then $(\nabla_X Y)(p)$ measures the rate of change of the wind vector observed along the motion: move from p in the direction $X(p)$, parallel-transport the wind vector back to T_pS^2, and differentiate.

In particular, if $Y = X$, then $\nabla_X X$ is the intrinsic acceleration of the motion. A curve c is a geodesic iff $\nabla_{c'} c' = 0$, i.e. it has zero intrinsic acceleration.

Example on the Earth sphere (equator): velocity field X and wind field Y

Model the Earth as the unit sphere $S^2 \subset \mathbb{R}^3$ with the induced (round) metric. Fix a point on the equator, for instance

$$p = (1, 0, 0) \in S^2.$$

Then the tangent plane at p is

$$T_pS^2 = \{v \in \mathbb{R}^3 : \ v \cdot p = 0\} = \{(0, a, b) : a, b \in \mathbb{R}\}.$$

A convenient orthonormal basis of T_pS^2 is given by

$$E(p) = (0, 1, 0) \qquad \text{(east, along the equator)}, \qquad N(p) = (0, 0, 1) \qquad \text{(north, toward the north pole)}.$$

Equator curve and "right/left" approach. Consider the equator (a great circle) through p:

$$c(t) = (\cos t,\ \sin t,\ 0), \qquad c(0) = p, \qquad c'(0) = (0, 1, 0) = E(p).$$

Approaching p from $t > 0$ versus $t < 0$ corresponds to approaching from opposite sides along the equator (east vs. west). If Y is a tangent vector field, then reversing direction flips the sign:

$$(\nabla_{-E}Y)(p) = -(\nabla_E Y)(p).$$

Why parallel transport is needed. For $t \neq 0$ one has $Y(c(t)) \in T_{c(t)}S^2$, while $Y(p) \in T_pS^2$, so the difference $Y(c(t)) - Y(p)$ is not intrinsic. Let $P_t : T_pS^2 \to T_{c(t)}S^2$ denote parallel transport along c. Then the covariant derivative (directional derivative of a tangent field) is

$$(\nabla_E Y)(p) = \frac{d}{dt}\bigg|_{t=0} \Big(P_t^{-1}\big(Y(c(t))\big)\Big) \in T_pS^2.$$

Interpretation: bring the wind vector back to the fixed tangent plane T_pS^2 without twisting, then differentiate there.

Ambient formula on the sphere (projection of the Euclidean derivative). Viewing Y as a vector-valued function on S^2 with values in $\mathbb{R}^3$ and extending it smoothly to a neighbourhood in $\mathbb{R}^3$, one has on S^2:

$$\nabla_X Y = \Pi_{T_pS^2}\big(D_X Y\big),$$

where $D_X Y$ is the usual directional derivative in $\mathbb{R}^3$ and

$$\Pi_{T_pS^2}(v) = v - (v \cdot p)\, p$$

is the orthogonal projection onto T_pS^2.

Two guiding interpretations.

- If $Y = X$, then $\nabla_X X$ is the *intrinsic acceleration*. A curve c is a geodesic iff $\nabla_{c'}c' = 0$. In particular, the equator is a great circle, hence a geodesic, so the velocity field along it satisfies $\nabla_E E = 0$.
- If Y represents a wind field (e.g. "always pointing north" along the equator), then $\nabla_E Y$ measures how that wind vector changes as you move east, after correcting for the rotation of tangent planes by parallel transport.

Parallel transport as a section of the restricted tangent bundle

Let M be a surface (e.g. S^2) and let $c : [0,T] \to M$ be a smooth curve. For each t the tangent space $T_{c(t)}M$ is a different vector space. The collection of all these tangent spaces along the curve is the restriction (pullback) of the tangent bundle to c:

$$c^*(TM) = \{(t,v) : t \in [0,T],\ v \in T_{c(t)}M\}.$$

One may think of $c^*(TM)$ as a "slice" of the tangent bundle TM along the curve, i.e. the family of tangent planes $\{T_{c(t)}M\}_{t\in[0,T]}$ stacked over the interval $[0,T]$.

A **vector field along** c is precisely a **section** of this restricted bundle:

$$V : [0,T] \to TM, \qquad V(t) \in T_{c(t)}M \quad \text{for all } t.$$

Let ∇ be the Levi–Civita connection on M. Given an initial vector $v \in T_{c(0)}M$, there exists a unique vector field $V(t)$ along c such that

$$V(0) = v, \qquad \nabla_{c'(t)}V(t) = 0 \quad \text{for all } t,$$

and this V is called the **parallel vector field** along c with initial value v.

Definition (parallel transport). The parallel transport along c from 0 to t is the linear isomorphism

$$P_{0\to t} : T_{c(0)}M \longrightarrow T_{c(t)}M, \qquad P_{0\to t}(v) := V(t),$$

where V is the unique parallel vector field along c satisfying $V(0) = v$. More generally, for $s,t \in [0,T]$ one writes $P_{s\to t} : T_{c(s)}M \to T_{c(t)}M$, with

$$P_{t\to t} = \mathrm{Id}, \qquad P_{s\to u} = P_{t\to u} \circ P_{s\to t}.$$

Problem 4 (Geodesic equations in local coordinates)

Show that the equations for geodesics on a smooth surface may be written locally in terms of coordinates $(u(t), v(t))$ as

$$\frac{d}{dt}\big(E\dot u + F\dot v\big) = \frac{1}{2}\big(E_u\dot u^2 + 2F_u\dot u\dot v + G_u\dot v^2\big), \qquad \frac{d}{dt}\big(F\dot u + G\dot v\big) = \frac{1}{2}\big(E_v\dot u^2 + 2F_v\dot u\dot v + G_v\dot v^2\big).$$

Solution. Let $x = x(u,v)$ be a local parametrisation and write the first fundamental form as

$$ds^2 = E\,du^2 + 2F\,du\,dv + G\,dv^2, \qquad E = \langle x_u, x_u\rangle,\ F = \langle x_u, x_v\rangle,\ G = \langle x_v, x_v\rangle.$$

For a curve $\gamma(t) = x(u(t), v(t))$ consider the energy functional

$$\mathcal{E} = \frac{1}{2}\int \big(E\dot u^2 + 2F\dot u\dot v + G\dot v^2\big)\,dt,$$

with Lagrangian

$$L(u,v,\dot u,\dot v) = \frac{1}{2}\big(E\dot u^2 + 2F\dot u\dot v + G\dot v^2\big).$$

The Euler–Lagrange equations are

$$\frac{d}{dt}\Big(\frac{\partial L}{\partial \dot u}\Big) = \frac{\partial L}{\partial u}, \qquad \frac{d}{dt}\Big(\frac{\partial L}{\partial \dot v}\Big) = \frac{\partial L}{\partial v}.$$

Compute

$$\frac{\partial L}{\partial \dot u} = E\dot u + F\dot v, \qquad \frac{\partial L}{\partial \dot v} = F\dot u + G\dot v,$$

and

$$\frac{\partial L}{\partial u} = \frac{1}{2}\big(E_u\dot u^2 + 2F_u\dot u\dot v + G_u\dot v^2\big), \qquad \frac{\partial L}{\partial v} = \frac{1}{2}\big(E_v\dot u^2 + 2F_v\dot u\dot v + G_v\dot v^2\big).$$

Substituting into Euler–Lagrange gives exactly the required identities. □

Problem 5 (No compact minimal surfaces in $\mathbb{R}^3$)

Show that there are no compact minimal surfaces in $\mathbb{R}^3$.

Solution. Let $X : S \to \mathbb{R}^3$ be an immersion of a surface S with unit normal N and mean curvature H. A standard identity is

$$\Delta_S X = 2H\,N,$$

where Δ_S is the Laplace–Beltrami operator on S. If S is minimal, then $H \equiv 0$, hence $\Delta_S X = 0$. Therefore each coordinate function x_1, x_2, x_3 of $X = (x_1, x_2, x_3)$ is harmonic on S.

If S is compact and without boundary, every harmonic function is constant (maximum principle, or $\int_S |\nabla x_i|^2 = -\int_S x_i \Delta_S x_i = 0$). Hence x_1, x_2, x_3 are constant on S, so $X(S)$ is a single point. This contradicts that X is an immersion. Therefore no compact minimal surface exists in $\mathbb{R}^3$. □

Minimal surfaces (quick theory)

Let $S \subset \mathbb{R}^3$ be a smooth oriented surface with unit normal N and principal curvatures k_1, k_2. The *mean curvature* is

$$H = \frac{k_1 + k_2}{2}.$$

The surface S is called *minimal* if

$$H \equiv 0 \qquad \text{(equivalently } k_1 + k_2 = 0\text{)}.$$

Variational characterization. Let X be a local parametrization of S and consider a normal variation

$$X_\varepsilon = X + \varepsilon\,\varphi N,$$

where φ is smooth with compact support in the interior. Then the first variation of area satisfies

$$\frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} \mathrm{Area}(S_\varepsilon) = -2\int_S H\,\varphi\,dA.$$

Hence S is minimal ($H \equiv 0$) iff it is a critical point of the area functional for all such variations.

Graph form. If S is given as a graph $z = u(x, y)$, then S is minimal iff u satisfies

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right) = 0,$$

equivalently

$$(1+u_y^2)u_{xx} - 2u_x u_y u_{xy} + (1+u_x^2)u_{yy} = 0.$$

Examples. Planes, the catenoid, the helicoid, and Enneper's surface.

Minimal surfaces (extended quick theory + examples/counterexamples)

Let $S \subset \mathbb{R}^3$ be a smooth oriented surface with unit normal N. At each point, the shape operator has eigenvalues k_1, k_2 (principal curvatures). The *mean curvature* is

$$H = \frac{k_1 + k_2}{2}.$$

The surface S is called *minimal* if

$$H \equiv 0 \qquad (\text{equivalently } k_1 + k_2 = 0).$$

Geometrically: the normal curvature in one principal direction is the negative of the other.

First variation of area (variational characterization). Let $X : U \to \mathbb{R}^3$ parametrize S and consider a normal variation

$$X_\varepsilon = X + \varepsilon\, \varphi N,$$

where φ is smooth and compactly supported in the interior. Then

$$\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} \operatorname{Area}(S_\varepsilon) = -2 \int_S H\, \varphi\, dA.$$

Hence S is minimal iff it is a critical point of area for all such normal variations.

Second variation (stability, optional but useful). If S is minimal, then the second variation has the form

$$\left.\frac{d^2}{d\varepsilon^2}\right|_{\varepsilon=0} \operatorname{Area}(S_\varepsilon) = \int_S \Big(|\nabla\varphi|^2 - (|A|^2 + \operatorname{Ric}(N, N))\varphi^2\Big)\, dA.$$

In $\mathbb{R}^3$ we have $\operatorname{Ric}(N, N) = 0$, so

$$\delta^2 A(\varphi) = \int_S \left(|\nabla\varphi|^2 - |A|^2\varphi^2\right)\, dA.$$

A minimal surface is *stable* if $\delta^2 A(\varphi) \ge 0$ for all compactly supported φ.

Graph form (minimal surface equation). If S is given as a graph $z = u(x, y)$ (upward orientation), then

$$2H = \operatorname{div}\left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right).$$

Thus S is minimal iff

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1+|\nabla u|^2}}\right) = 0,$$

equivalently

$$(1+u_y^2)u_{xx} - 2u_xu_yu_{xy} + (1+u_x^2)u_{yy} = 0.$$

For *small slopes* $|\nabla u| \ll 1$, the equation linearizes to $\Delta u \approx 0$, so minimal graphs are a nonlinear analogue of harmonic functions.

Harmonic coordinate functions (very handy fact). A surface S is minimal iff each coordinate function of the immersion $X = (x^1, x^2, x^3)$ is harmonic with respect to the induced metric, i.e.

$$\Delta_S x^i = 0 \quad (i = 1, 2, 3).$$

Equivalently, the mean curvature vector satisfies $\vec{H} = HN \equiv 0$.

Maximum principle (intuition). A minimal surface cannot have an *interior* point where it lies strictly on one side of a tangent plane (unless it is locally that plane). This is a strong rigidity tool for proving uniqueness and nonexistence.

Examples (minimal, $H \equiv 0$).

- **Planes:** $k_1 = k_2 = 0$ hence $H = 0$.
- **Catenoid (surface of revolution):**

  $$X(u, v) = (\cosh v \cos u,\ \cosh v \sin u,\ v).$$

  It is minimal and is a classical soap-film shape between two circles.
- **Helicoid (ruled surface):**

  $$X(u, v) = (v \cos u,\ v \sin u,\ au),$$

  minimal for any constant pitch $a \neq 0$.
- **Enneper surface (minimal, self-intersecting):**

  $$X(u, v) = \left(u - \frac{u^3}{3} + uv^2,\ v - \frac{v^3}{3} + vu^2,\ u^2 - v^2\right).$$

- **Scherk minimal graph (explicit PDE solution):**

  $$u(x, y) = \log\left(\frac{\cos y}{\cos x}\right) \quad \text{(on a domain where } \cos x, \cos y > 0\text{)}.$$

Counterexamples (not minimal, $H \not\equiv 0$).

- **Sphere of radius R:** $k_1 = k_2 = \frac{1}{R}$, so $H = \frac{1}{R} \neq 0$ (constant mean curvature).
- **Circular cylinder of radius R:** $k_1 = \frac{1}{R}$, $k_2 = 0$, so $H = \frac{1}{2R} \neq 0$.
- **Paraboloid $z = x^2 + y^2$:** a graph with strictly positive mean curvature (see the H-heatmap chart).

- **Cone:** not smooth at the apex; away from the apex it is generally not minimal (and minimality is a smooth concept).

Charts (included files).

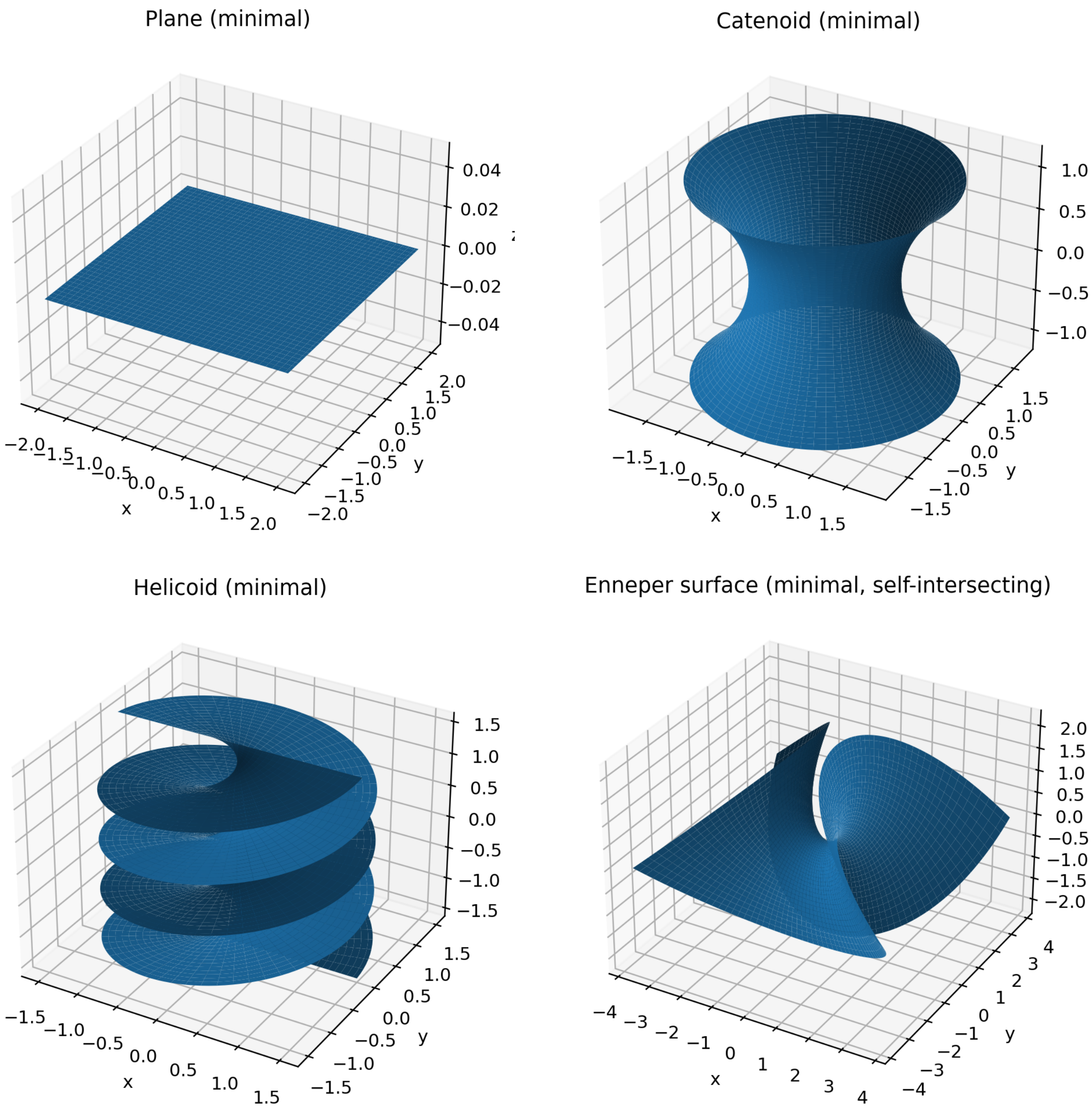


Figure 7: Classic minimal surfaces: plane, catenoid, helicoid, Enneper.

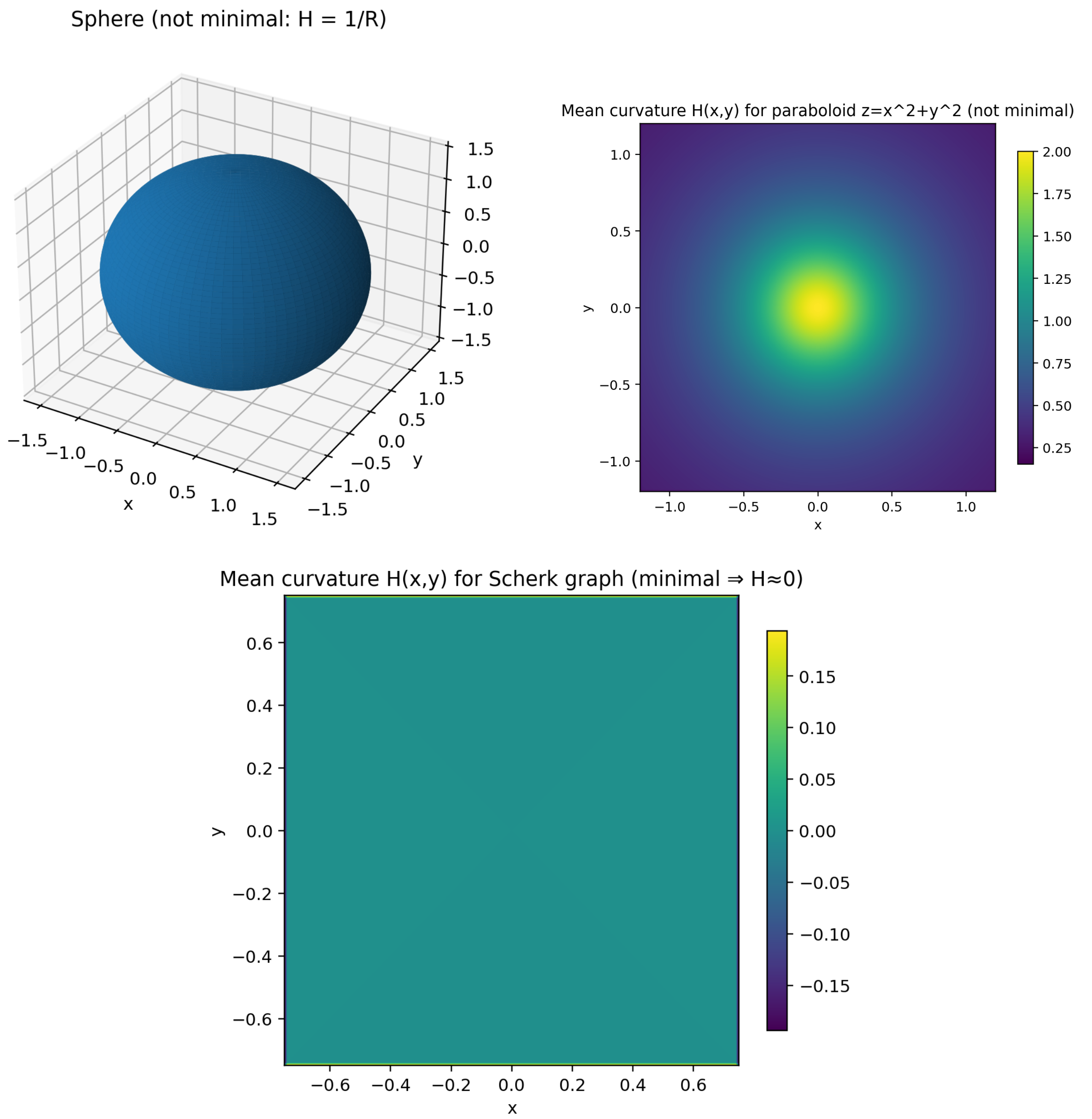


Figure 8: Counterexamples vs example via mean curvature: sphere has constant $H \neq 0$; paraboloid has $H > 0$ varying; Scherk graph has $H \approx 0$ (numerically).

Harmonic coordinate functions (what it means). Let $X\colon U \to \mathbb{R}^3$ be a smooth immersion whose image is a surface $S = X(U)$. Write local coordinates (u^1, u^2) on U and denote

$$X_i = \frac{\partial X}{\partial u^i} \qquad (i = 1, 2).$$

The *induced metric* (first fundamental form) on S is the 2×2 matrix

$$g_{ij} = \langle X_i, X_j \rangle, \qquad g = \det(g_{ij}), \qquad (g^{ij}) = (g_{ij})^{-1}.$$

Laplace–Beltrami operator. For a smooth function $f\colon S \to \mathbb{R}$ (locally $f = f(u^1, u^2)$), the intrinsic Laplacian on S is

$$\Delta_S f = \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^i} \left(\sqrt{g}\, g^{ij} \frac{\partial f}{\partial u^j} \right),$$

(Einstein summation over $i, j = 1, 2$). We say f is *harmonic on S* if $\Delta_S f = 0$.

Coordinate functions. Let (x^1, x^2, x^3) be the standard coordinates on $\mathbb{R}^3$. Their restrictions to the surface are the functions

$$x^k|_S = x^k \circ X, \qquad k = 1, 2, 3.$$

Saying "each coordinate function is harmonic" means

$$\Delta_S(x^k \circ X) = 0 \qquad (k = 1, 2, 3).$$

Vector form and mean curvature vector. We can apply Δ_S componentwise to the immersion $X = (x^1 \circ X,\ x^2 \circ X,\ x^3 \circ X)$ and define

$$\Delta_S X := \big(\Delta_S(x^1 \circ X),\ \Delta_S(x^2 \circ X),\ \Delta_S(x^3 \circ X)\big) \in \mathbb{R}^3.$$

A fundamental identity (convention-dependent by a sign/factor) is

$$\Delta_S X = -2H\, N,$$

where N is the unit normal and H is the mean curvature used in the first variation formula $\frac{d}{d\varepsilon}\big|_0 \mathrm{Area}(S_\varepsilon) = -2\int_S H\varphi\, dA$.

Conclusion (equivalence). From $\Delta_S X = -2H\, N$ we get

$$\Delta_S(x^k \circ X) = 0\ \forall k \quad \Longleftrightarrow \quad \Delta_S X = 0 \quad \Longleftrightarrow \quad H \equiv 0.$$

Equivalently, the *mean curvature vector* $\vec{H}$ (often defined as $\vec{H} := HN$ or $\vec{H} := 2HN$, depending on convention) vanishes:

$$\vec{H} = 0.$$

Interpretation. Thus a minimal surface is exactly a surface whose inclusion map X is a *harmonic map* (with respect to the induced metric): its coordinate functions have no intrinsic "source term" on the surface. "Harmonic" here is intrinsic (depends on the geometry of S), not the flat Laplacian in $\mathbb{R}^2$.

Quick check (example vs counterexample).

- Plane: in parameters $X(u, v) = (u, v, 0)$ the induced metric is Euclidean, so Δ_S is the usual Laplacian and $\Delta_S X = 0$, hence $H = 0$.
- Sphere of radius R: one has $\Delta_S X = -\frac{2}{R^2} X \neq 0$, so the coordinate functions are not harmonic and the sphere is not minimal (indeed $H = \frac{1}{R}$).

In plain words.

- Δ_S is the *intrinsic* Laplacian on the surface (the Laplace–Beltrami operator). It measures how a function bends *along the surface*, using the surface's metric.
- The functions x, y, z (restricted to the surface) being *harmonic* means:

$$\Delta_S x = \Delta_S y = \Delta_S z = 0.$$

- The identity

$$\Delta_S X = -2HN$$

  tells you that the "intrinsic Laplacian of the position vector" points in the normal direction, and its size is controlled by the mean curvature.
- Therefore:

$$S \text{ is minimal } \Longleftrightarrow H \equiv 0 \Longleftrightarrow \Delta_S X = 0 \Longleftrightarrow \Delta_S x = \Delta_S y = \Delta_S z = 0.$$

Derivation sketch of $\Delta_S X = -2HN$ (few lines). Let $X\colon U \to \mathbb{R}^3$ be a local parametrization with coordinates (u^1, u^2), tangent vectors $X_i = \partial_i X$, induced metric $g_{ij} = \langle X_i, X_j\rangle$, and Laplace–Beltrami operator

$$\Delta_S f = \frac{1}{\sqrt{g}}\partial_i\big(\sqrt{g}\, g^{ij}\partial_j f\big)\,.$$

Apply Δ_S componentwise to X:

$$\Delta_S X = \frac{1}{\sqrt{g}}\partial_i\big(\sqrt{g}\, g^{ij} X_j\big) = g^{ij}\big(\partial_i X_j - \Gamma^k_{ij} X_k\big).$$

By the Gauss formula (decomposition of second derivatives),

$$\partial_i X_j = \Gamma^k_{ij} X_k + b_{ij} N,$$

so

$$\Delta_S X = g^{ij} b_{ij}\, N.$$

Since $H = \frac{1}{2}\,\mathrm{tr}_g(b) = \frac{1}{2} g^{ij} b_{ij}$, we obtain

$$\Delta_S X = (2H)\, N.$$

Convention note. Many texts define b_{ij} (or H) with the opposite sign (depending on whether $b_{ij} = \langle \partial_i X_j, N\rangle$ or $b_{ij} = -\langle \partial_i X_j, N\rangle$, and on the orientation of N). With the convention used in the first-variation formula $\frac{d}{d\varepsilon}\big|_0 \mathrm{Area}(S_\varepsilon) = -2\int_S H\varphi\, dA$, this identity is written as

$$\Delta_S X = -2HN.$$

Concrete calculations (2 examples).

Example 1 (minimal): plane. Parametrize the plane by

$$X(u, v) = (u, v, 0).$$

Then $X_u = (1, 0, 0)$, $X_v = (0, 1, 0)$, hence $g_{ij} = \delta_{ij}$, $\sqrt{g} = 1$, and

$$\Delta_S = \partial_{uu} + \partial_{vv}.$$

Therefore

$$\Delta_S X = (\Delta_S u,\ \Delta_S v,\ \Delta_S 0) = (0, 0, 0).$$

So $\Delta_S X = 0$, hence $H = 0$: the plane is minimal.

Example 2 (not minimal): sphere of radius R. Use spherical coordinates (θ, φ):

$$X(\theta, \varphi) = \big(R\sin\theta\cos\varphi,\ R\sin\theta\sin\varphi,\ R\cos\theta\big).$$

The induced metric is

$$(g_{ij}) = \begin{pmatrix} R^2 & 0 \\ 0 & R^2\sin^2\theta \end{pmatrix},$$

so the Laplace–Beltrami operator on S^2_R is

$$\Delta_S f = \frac{1}{R^2}\left(\frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta f) + \frac{1}{\sin^2\theta}\partial_{\varphi\varphi} f\right).$$

Let $x(\theta,\varphi) = R\sin\theta\cos\varphi$ (the first coordinate). Then

$$\partial_\theta x = R\cos\theta\cos\varphi, \qquad \partial_\varphi x = -R\sin\theta\sin\varphi,$$

$$\partial_{\varphi\varphi} x = -R\sin\theta\cos\varphi = -x,$$

and

$$\frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta x) = \frac{1}{\sin\theta}\partial_\theta\big(R\sin\theta\cos\theta\cos\varphi\big) = \frac{R(\cos^2\theta - \sin^2\theta)\cos\varphi}{\sin\theta}.$$

Putting this together,

$$\Delta_S x = \frac{1}{R^2}\left(\frac{R(\cos^2\theta-\sin^2\theta)\cos\varphi}{\sin\theta} + \frac{1}{\sin^2\theta}(-x)\right) = -\frac{2}{R^2}\,x.$$

Similarly,

$$\Delta_S y = -\frac{2}{R^2}\,y, \qquad \Delta_S z = -\frac{2}{R^2}\,z,$$

hence (vectorially)

$$\Delta_S X = -\frac{2}{R^2}X \neq 0.$$

Now $N = \frac{1}{R}X$ (outward unit normal) and the sphere has $H = \frac{1}{R}$, so

$$-2HN = -2\cdot\frac{1}{R}\cdot\frac{1}{R}X = -\frac{2}{R^2}X,$$

which matches the computed $\Delta_S X$. Therefore the sphere is *not* minimal.

Takeaway: on the plane, the restricted coordinates are harmonic; on the sphere they are eigenfunctions with eigenvalue $-2/R^2$, so they are not harmonic and $H \neq 0$.

Continuing the sphere example: the 2nd and 3rd coordinates. For the sphere parametrization

$$X(\theta,\varphi) = \big(R\sin\theta\cos\varphi,\ R\sin\theta\sin\varphi,\ R\cos\theta\big),$$

we already computed for the first coordinate $x(\theta,\varphi) = R\sin\theta\cos\varphi$ that

$$\Delta_S x = -\frac{2}{R^2}\,x.$$

The same should be done for the other two coordinate functions.

Second coordinate. Let $y(\theta,\varphi) = R\sin\theta\sin\varphi$. Then

$$\partial_\theta y = R\cos\theta\sin\varphi, \qquad \partial_{\varphi\varphi} y = -R\sin\theta\sin\varphi = -y,$$

and

$$\frac{1}{\sin\theta}\partial_\theta\big(\sin\theta\,\partial_\theta y\big) = \frac{1}{\sin\theta}\partial_\theta\big(R\sin\theta\cos\theta\sin\varphi\big) = \frac{R(\cos^2\theta-\sin^2\theta)\sin\varphi}{\sin\theta}.$$

Substituting into

$$\Delta_S f = \frac{1}{R^2}\left(\frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta f) + \frac{1}{\sin^2\theta}\partial_{\varphi\varphi} f\right),$$

we obtain

$$\Delta_S y = \frac{1}{R^2}\left(\frac{R(\cos^2\theta - \sin^2\theta)\sin\varphi}{\sin\theta} + \frac{1}{\sin^2\theta}(-y)\right) = -\frac{2}{R^2}\,y.$$

Third coordinate. Let $z(\theta,\varphi) = R\cos\theta$. Then $\partial_\varphi z = 0$ and

$$\partial_\theta z = -R\sin\theta, \qquad \sin\theta\,\partial_\theta z = -R\sin^2\theta, \qquad \partial_\theta(\sin\theta\,\partial_\theta z) = -2R\sin\theta\cos\theta.$$

Hence

$$\frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta z) = -2R\cos\theta = -2z,$$

and therefore

$$\Delta_S z = \frac{1}{R^2}\,(-2z+0) = -\frac{2}{R^2}\,z.$$

Vector form. Since $X = (x,y,z)$, applying Δ_S componentwise gives

$$\Delta_S X = \big(\Delta_S x, \Delta_S y, \Delta_S z\big) = -\frac{2}{R^2}(x,y,z) = -\frac{2}{R^2}X \neq 0.$$

Thus the coordinate functions are *not* harmonic on the sphere, and $H \neq 0$ (in fact $H = \frac{1}{R}$ with outward normal $N = \frac{1}{R}X$, so $\Delta_S X = -2HN$).

Variational meaning of minimal surfaces. Let $S \subset \mathbb{R}^3$ be a smooth oriented surface with unit normal field N and mean curvature H. Fix the boundary (if present) and consider normal variations supported in the interior:

$$X_\varepsilon = X + \varepsilon\,\varphi N, \qquad \varphi \in C_c^\infty(\operatorname{int} S).$$

Being a *critical point* of the area functional means

$$\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} \operatorname{Area}(S_\varepsilon) = 0 \quad \text{for all such } \varphi.$$

The first variation formula states

$$\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} \operatorname{Area}(S_\varepsilon) = -2\int_S H\,\varphi\,dA.$$

Hence the critical point condition for all test functions φ is equivalent to

$$H \equiv 0.$$

Therefore:

$$S \text{ is minimal} \quad \Longleftrightarrow \quad S \text{ is a critical point of the area functional.}$$

Example: the catenoid as a minimal surface. A standard parametrization of the catenoid is

$$X(u,v) = \big(a\cosh(v/a)\cos u,\ a\cosh(v/a)\sin u,\ v\big), \qquad u \in [0,2\pi),\ v \in \mathbb{R},$$

where $a > 0$ is a constant. This surface satisfies $H \equiv 0$ everywhere, hence it is minimal.

Remark (critical point vs minimum). The condition $H \equiv 0$ guarantees only that the first variation vanishes (critical point). Whether the surface is a *local minimum* depends on the second variation (stability), and a minimal surface need not be globally area-minimizing.

What are "restricted coordinates"? Let (x^1, x^2, x^3) be the standard coordinate functions on $\mathbb{R}^3$:

$$x^1(x,y,z) = x, \qquad x^2(x,y,z) = y, \qquad x^3(x,y,z) = z.$$

If $S \subset \mathbb{R}^3$ is a surface and $i : S \hookrightarrow \mathbb{R}^3$ is the inclusion, then the *restricted coordinate functions* on S are

$$x^k|_S := x^k \circ i \ :\ S \to \mathbb{R} \qquad (k = 1,2,3).$$

Equivalently, if $X : U \to S$ is a parametrization with $X(u,v) = (X^1(u,v), X^2(u,v), X^3(u,v))$, then

$$x^k|_S \circ X = x^k \circ X = X^k(u,v),$$

so the restricted coordinates are exactly the component functions of the immersion X.

Problem 6 (Minimal surfaces and conformality of the Gauss map)

Let S be a regular surface without umbilical points. Prove that S is minimal iff the Gauss map $N : S \to S^2$ satisfies

$$\langle dN_p(v_1), dN_p(v_2)\rangle = \lambda(p)\,\langle v_1, v_2\rangle$$

for all $p \in S$ and all $v_1, v_2 \in T_pS$, where $\lambda(p) \neq 0$ depends only on p. By considering stereographic projection deduce that isothermal coordinates exist around a nonplanar point in a minimal surface.

Solution. Let $S_p : T_pS \to T_pS$ be the shape operator. The Weingarten relation is

$$dN_p(v) = -S_p(v).$$

Thus the condition becomes

$$\langle S_p(v_1), S_p(v_2)\rangle = \lambda(p)\,\langle v_1, v_2\rangle \qquad (\lambda(p) \neq 0).$$

Choose an orthonormal principal basis (e_1, e_2) at p, so $S_p(e_i) = k_i e_i$ where k_1, k_2 are the principal curvatures. Writing $v = a_1e_1 + a_2e_2$ and $w = b_1e_1 + b_2e_2$,

$$\langle S_p(v), S_p(w)\rangle = k_1^2 a_1b_1 + k_2^2 a_2b_2, \qquad \langle v, w\rangle = a_1b_1 + a_2b_2.$$

Since the identity holds for all v, w, it follows that $k_1^2 = k_2^2 = \lambda(p)$. Because S has no umbilics we have $k_1 \neq k_2$, hence $k_2 = -k_1 \neq 0$. Therefore

$$H(p) = \frac{k_1 + k_2}{2} = 0,$$

so S is minimal.

Conversely, if S is minimal and has no umbilics, then $k_2 = -k_1 \neq 0$, hence $k_1^2 = k_2^2$ and the identity holds with $\lambda(p) = k_1(p)^2 = -K(p) > 0$.

For the isothermal coordinates: on a minimal surface (without umbilics at the point considered) the Gauss map N is conformal, and at a nonplanar point p we have $dN_p \neq 0$. Stereographic projection $\pi : S^2 \setminus \{\text{north pole}\} \to \mathbb{C}$ is conformal, hence $g = \pi \circ N$ is conformal. Since $dN_p \neq 0$, g is a local diffeomorphism near p. Writing $g = u + iv$, the coordinates (u, v) are isothermal, i.e.

$$ds^2 = \rho(u, v)^2(du^2 + dv^2)$$

for some positive function ρ. □

Where does dN_p come from? The Gauss map is $N : S \to S^2$, $N(p) =$ unit normal at p. Its differential at p is the linear map

$$dN_p : T_pS \to T_{N(p)}S^2,$$

defined by: for $v \in T_pS$, choose a smooth curve $\gamma(t) \subset S$ with $\gamma(0) = p$ and $\gamma'(0) = v$, and set

$$dN_p(v) = \left.\frac{d}{dt}\right|_0 N(\gamma(t)).$$

Since $\|N(\gamma(t))\|^2 = 1$, differentiating gives $\langle N(p), dN_p(v)\rangle = 0$, hence $dN_p(v) \in T_{N(p)}S^2$. Moreover, the shape operator is $S_p = -dN_p$ (via the natural identification of tangent planes in $\mathbb{R}^3$), so dN_p encodes the principal curvatures.

Two normals and orientability. At each $p \in S \subset \mathbb{R}^3$ the normal line is 1-dimensional, hence there are exactly two unit normals: $N(p)$ and $-N(p)$. A *Gauss map* is a choice of one of these at every point, i.e. a map $N : S \to S^2$ with $N(p) \perp T_pS$.

A surface is *orientable* iff one can choose the unit normal *continuously* (equivalently smoothly) on all of S, i.e. iff there exists a global smooth Gauss map $N : S \to S^2$. Non-orientability means that local choices of normal exist on charts but necessarily flip sign on some overlaps/loops.

If $X(u, v)$ is a local parametrization, a local unit normal is

$$N = \frac{X_u \times X_v}{\|X_u \times X_v\|}.$$

On an overlap of two charts, the corresponding normals differ by a sign given by the Jacobian of the coordinate change: the normal flips when the change of coordinates reverses orientation.

Finally, choosing the opposite normal replaces N by $-N$ and dN_p by $-dN_p$, but

$$\langle dN_p(v_1), dN_p(v_2)\rangle$$

is unchanged, so conformality statements are independent of the sign choice.

Problem 7: Weierstrass data: immersion criterion

Let $D \subset \mathbb{C}$, and f and g be functions on D giving a Weierstrass representation of a parametrisation ϕ. Show that ϕ is an immersion iff f vanishes only at the poles of g and the order of its zero at such a point is *exactly twice* the order of the pole of g.

Solution. In Weierstrass form one may write (up to an overall constant depending on convention)

$$\phi_z = \big((1-g^2)f,\ i(1+g^2)f,\ 2gf\big),$$

a holomorphic $\mathbb{C}^3$-valued 1-form. The map is an immersion iff $\phi_z \neq 0$ everywhere.

If g is holomorphic at p and $f(p) = 0$, then all components of ϕ_z vanish at p, so ϕ is not an immersion. Hence f may vanish only at poles of g.

Now let p be a pole of g of order m. Choose a local coordinate z with $z(p) = 0$ and write

$$g(z) = az^{-m} + \cdots, \quad a \neq 0; \qquad f(z) = bz^n + \cdots, \quad b \neq 0.$$

Then

$$(1-g^2)f \sim (-a^2)b\, z^{n-2m}, \qquad (1+g^2)f \sim (a^2)b\, z^{n-2m}, \qquad 2gf \sim 2ab\, z^{n-m}.$$

For ϕ_z to be holomorphic at p we need $n - 2m \geq 0$, i.e. $n \geq 2m$. If $n > 2m$, then each component vanishes at $z = 0$, hence $\phi_z(p) = 0$ and ϕ fails to be an immersion. Thus we must have $n = 2m$, and in that case $(1-g^2)f$ has a nonzero constant term, so $\phi_z(p) \neq 0$. Therefore ϕ is an immersion iff f vanishes only at poles of g and $\mathrm{ord}_p(f) = 2\,\mathrm{ord}_p(\text{pole of } g)$. $\square$

Weierstrass–Enneper representation (parametrization of minimal surfaces)

Goal. Describe all (simply connected) minimal surfaces in $\mathbb{R}^3$ using holomorphic data.

Setup. Let $\Omega \subset \mathbb{C}$ be simply connected with complex coordinate $z = u + iv$. A map $X : \Omega \to \mathbb{R}^3$ is called a *conformal parametrization* if the induced metric has the form

$$ds^2 = e^{2\omega}(du^2 + dv^2) \quad (\text{equivalently } \langle X_u, X_v\rangle = 0,\ |X_u| = |X_v|).$$

For conformal parametrizations, the minimality condition $H \equiv 0$ becomes equivalent to harmonicity of X:

$$H \equiv 0 \quad \Longleftrightarrow \quad X_{uu} + X_{vv} = 0 \quad \Longleftrightarrow \quad X \text{ is harmonic componentwise.}$$

Weierstrass data. Choose

$$g : \Omega \to \widehat{\mathbb{C}} \quad (\text{meromorphic}), \qquad \eta = h(z)\, dz \quad (\text{holomorphic 1-form}),$$

such that zeros of η cancel poles/zeros of g so that the expressions below are holomorphic.

Define holomorphic 1-forms

$$\phi_1 = \frac{1}{2}\left(\frac{1}{g} - g\right)\eta, \qquad \phi_2 = \frac{i}{2}\left(\frac{1}{g} + g\right)\eta, \qquad \phi_3 = \eta.$$

They satisfy the key identity

$$\phi_1^2 + \phi_2^2 + \phi_3^2 = 0.$$

Representation theorem. Define

$$X(z) = \Re \int^z (\phi_1, \phi_2, \phi_3).$$

Then $X : \Omega \to \mathbb{R}^3$ is a *conformal minimal immersion* (away from points where all ϕ_k vanish). Conversely, every conformal minimal immersion on a simply connected domain arises this way.

Meaning of g and η.

- g is the *stereographic projection of the Gauss map*: if $N : \Omega \to S^2$ is the unit normal, then $g = \mathrm{stereo}(N)$.
- η controls the conformal factor (size) of the metric.

Induced metric (useful formula). The first fundamental form becomes

$$ds^2 = \big(|\phi_1|^2 + |\phi_2|^2 + |\phi_3|^2\big) = \frac{(1+|g|^2)^2}{4|g|^2}\,|\eta|^2,$$

so the parametrization is conformal by construction.

Associate family (catenoid $\leftrightarrow$ helicoid). If (g, η) gives a minimal surface, then for any $\alpha \in \mathbb{R}$,

$$(g,\ e^{i\alpha}\eta)$$

gives another minimal surface with the *same metric*; this produces the classical associate family.

Example 1: Enneper surface. Take

$$g(z) = z, \qquad \eta = dz.$$

Then

$$X(z) = \Re \int \left(\frac{1}{2}\left(\frac{1}{z} - z\right),\ \frac{i}{2}\left(\frac{1}{z} + z\right),\ 1 \right) dz,$$

which yields (after integration and taking real parts) the standard Enneper parametrization.

Example 2: Catenoid and helicoid (associate surfaces). Let

$$g(z) = e^z, \qquad \eta = dz.$$

Then

$$\phi_1 = \frac{1}{2}(e^{-z} - e^z)dz = -\sinh(z)\,dz, \qquad \phi_2 = \frac{i}{2}(e^{-z} + e^z)dz = i\cosh(z)\,dz, \qquad \phi_3 = dz.$$

Integrating and taking real parts gives

$$X(z) = \Re\big(-\cosh z,\ i\sinh z,\ z\big).$$

Writing $z = u + iv$,

$$X(u,v) = \big(-\cosh u \cos v,\ -\cosh u \sin v,\ u\big),$$

which is a catenoid (up to a rigid motion/sign change).

If instead we replace η by $i\,dz$ (i.e. $\alpha = \pi/2$ in the associate family), we obtain the helicoid (again up to rigid motion). This illustrates that catenoid and helicoid have the same intrinsic metric but differ extrinsically.

Problem 8 (Weierstrass data for catenoid and helicoid)

We use the Weierstrass representation in the (f,g)-form:

$$X(z) = \Re \int^z \Big(\tfrac{1}{2}f(1-g^2),\ \tfrac{i}{2}f(1+g^2),\ fg\Big)\,dz,$$

where g is meromorphic, f is holomorphic, and the resulting parametrization is conformal and minimal.

Choice of complex parameter. For the given real parameters (u,v), the catenoid/helicoid are conformal in (u,v), and it is convenient to set

$$z = v + iu.$$

Thus a suitable domain is the strip

$$D = \{z = v + iu \in \mathbb{C} :\ v \in \mathbb{R},\ 0 < u < 2\pi\}.$$

Catenoid. Take

$$g(z) = e^z, \qquad f(z) = ae^{-z}.$$

Then

$$\tfrac{1}{2}f(1-g^2) = \frac{a}{2}(e^{-z} - e^z) = -a\sinh z, \qquad \tfrac{i}{2}f(1+g^2) = \frac{ia}{2}(e^{-z} + e^z) = ia\cosh z,$$

and $fg = a$. Hence

$$X(z) = \Re\big(-a\cosh z,\ ia\sinh z,\ az\big).$$

With $z = v + iu$ this becomes (up to a rigid motion in the xy-plane)

$$X(u,v) = \big(a\cosh v\cos u,\ a\cosh v\sin u,\ av\big),$$

which is exactly the given catenoid parametrization (possibly after rotating by π about the z-axis).

Helicoid. Keep the same $g(z) = e^z$ but replace f by if:

$$g(z) = e^z, \qquad f(z) = iae^{-z}.$$

This is the classical associate surface construction. Substituting yields a parametrization which, after a rigid motion and a harmless shift of u, is the given helicoid

$$\phi(u,v) = \big(a\sinh v\cos u,\ a\sinh v\sin u,\ au\big).$$

(Geometrically: multiplying f by i rotates the Weierstrass integrand by 90° in the associate family.)

Problem 9 (Gaussian curvature formula and isolated zeros)

Let a minimal surface be given by Weierstrass data (f, g) as in Problem 8:

$$X(z) = \Re \int^z \Big(\tfrac{1}{2} f(1-g^2),\ \tfrac{i}{2} f(1+g^2),\ fg \Big)\, dz.$$

Step 1: induced metric. In the parameter $z = u + iv$ the parametrization is conformal and one has

$$ds^2 = \lambda^2 |dz|^2, \qquad \lambda = \frac{|f|(1+|g|^2)}{2},$$

equivalently

$$E = G = \lambda^2 = \frac{|f|^2(1+|g|^2)^2}{4}, \qquad F = 0.$$

Step 2: Gauss map and stereographic projection. For a minimal surface, the Gauss map $N : S \to S^2$ is conformal in conformal coordinates, and its stereographic coordinate is precisely g. The round metric on S^2 written in the stereographic coordinate g is

$$ds^2_{S^2} = \frac{4}{(1+|g|^2)^2} |dg|^2,$$

so the pullback satisfies

$$|N_u|^2 + |N_v|^2 = \frac{4}{(1+|g|^2)^2} \big(|g_u|^2 + |g_v|^2 \big).$$

Since g is holomorphic (meromorphic), $g_u = g'(z)$ and $g_v = ig'(z)$, hence

$$|g_u|^2 + |g_v|^2 = 2|g'(z)|^2,$$

and therefore

$$|N_u|^2 + |N_v|^2 = \frac{8|g'|^2}{(1+|g|^2)^2}.$$

Step 3: relate K to dN for minimal surfaces. In an orthonormal basis of T_pS, the eigenvalues of dN_p are $-k_1, -k_2$. For a minimal surface $k_2 = -k_1$, so $K = k_1 k_2 = -k_1^2 \le 0$ and

$$\|dN\|^2 = k_1^2 + k_2^2 = 2k_1^2 = -2K, \qquad \text{hence} \qquad K = -\frac{1}{2}\|dN\|^2.$$

In conformal coordinates, an orthonormal basis is $e_1 = X_u/\lambda$, $e_2 = X_v/\lambda$, so

$$\|dN\|^2 = \frac{|N_u|^2 + |N_v|^2}{\lambda^2}.$$

Thus

$$K = -\frac{1}{2} \frac{|N_u|^2 + |N_v|^2}{\lambda^2} = -\frac{1}{2} \frac{\frac{8|g'|^2}{(1+|g|^2)^2}}{\frac{|f|^2(1+|g|^2)^2}{4}} = -\frac{16|g'|^2}{|f|^2(1+|g|^2)^4}.$$

Equivalently,

$$K = -\left(\frac{4|g'|}{|f|(1+|g|^2)^2} \right)^2.$$

Zeros of K. Assume the surface is regular (no branch points), so $f \neq 0$. Then $K(p) = 0$ iff $g'(p) = 0$. But g' is holomorphic wherever g is holomorphic, hence either $g' \equiv 0$ or its zeros are isolated. If $g' \equiv 0$, then g is constant, the Gauss map is constant, and the surface is a plane, so $K \equiv 0$. Otherwise, the zeros of g' (hence of K) are isolated.

Problem 10. Associated family in Weierstrass data (local isometry)

Problem statement. The Weierstrass representation is not unique: if $\phi_{(f,g)} : D \to \mathbb{R}^3$ is the associated parametrization and $\alpha : W \to D$ is a bijective holomorphic map, then $\phi_{(f,g)} \circ \alpha$ is another representation of the same minimal surface and it must have the same form with different f and g (which should be specified). By choosing $\alpha(z) = g^{-1}(z)$, show that, locally around regular points of g (where $g' \neq 0$), we can assume that our pair (f, g) is of the form (F, id), for some local holomorphic function F. We denote such a representation by ϕ_F. Show that the minimal surfaces given by $\phi_{e^{-i\theta}F}$ for $\theta \in \mathbb{R}$ are all locally isometric.

Theory / setup

We use the Weierstrass data (f, g) (holomorphic on D, with g meromorphic in general) in the form

$$\phi_{(f,g)}(z) = \Re \int^z \Big(\tfrac{1}{2} f(1 - g^2),\ \tfrac{i}{2} f(1 + g^2),\ fg \Big)\, dz.$$

Its induced metric is

$$ds^2 = \frac{|f|^2(1 + |g|^2)^2}{4}\, |dz|^2.$$

Solution (core argument)

Let $\alpha : W \to D$ be biholomorphic and consider $\phi_{(f,g)} \circ \alpha$. Writing $z = \alpha(w)$, we have $dz = \alpha'(w)\, dw$, hence

$$\phi_{(f,g)}(\alpha(w)) = \Re \int^w \Big(\tfrac{1}{2} f(\alpha(\xi))(1 - g(\alpha(\xi))^2),\ \tfrac{i}{2} f(\alpha(\xi))(1 + g(\alpha(\xi))^2),\ f(\alpha(\xi))g(\alpha(\xi)) \Big)\, \alpha'(\xi)\, d\xi.$$

Therefore $\phi_{(f,g)} \circ \alpha$ is again a Weierstrass representation with new data

$$\tilde{g}(w) = g(\alpha(w)), \qquad \tilde{f}(w) = f(\alpha(w))\, \alpha'(w).$$

Now assume $g'(z_0) \neq 0$. By the holomorphic inverse function theorem, there is a neighborhood on which g admits a holomorphic inverse g^{-1}. Choose $\alpha = g^{-1}$. Then locally

$$\tilde{g}(w) = g(g^{-1}(w)) = w = \mathrm{id}(w), \qquad \tilde{f}(w) = f(g^{-1}(w))\, (g^{-1})'(w) =: F(w),$$

so the pair has the form (F, id). We denote the corresponding immersion by ϕ_F.

Finally, for $g(z) = z$ the induced metric becomes

$$ds^2 = \frac{|F(z)|^2(1 + |z|^2)^2}{4}\, |dz|^2.$$

Replacing F by $e^{-i\theta}F$ does not change $|F|$, hence produces the same metric. Thus the identity map on the parameter domain is a local isometry between ϕ_F and $\phi_{e^{-i\theta}F}$. Therefore the family $\{\phi_{e^{-i\theta}F}\}_{\theta \in \mathbb{R}}$ consists of locally isometric minimal surfaces.

Examples

1. **Catenoid–helicoid associated family (classical).** Take $g(z) = z$ and $f(z) = 1/z^2$ (on a punctured domain). Then $f_\theta(z) = e^{-i\theta}/z^2$ yields the associated family. In particular, $\theta = 0$ gives (a parametrization of) the catenoid and $\theta = \pi/2$ gives (a parametrization of) the helicoid, and they are locally isometric since the induced metric depends only on $|f_\theta| = |f|$.

2. **Enneper associated family.** Take $g(z) = z$ and $f(z) = 1$ (on a simply connected domain). Then $f_\theta = e^{-i\theta}$ produces the associated Enneper family, all locally isometric for the same reason.

Theory: Weierstrass data, metric, and reparametrization

Let $D \subset \mathbb{C}$ be simply connected and let (f, g) be holomorphic data (with g meromorphic in general). Define the $\mathbb{C}^3$-valued holomorphic 1-form

$$\Phi_{(f,g)}(z)\,dz = \Big(\tfrac{1}{2}f(1-g^2),\ \tfrac{i}{2}f(1+g^2),\ fg\Big)\,dz.$$

The associated minimal immersion is given by

$$\phi_{(f,g)}(z) = \Re \int^z \Phi_{(f,g)}(\zeta)\,d\zeta.$$

The choice of $\Phi_{(f,g)}$ ensures $\Phi_{(f,g)} \cdot \Phi_{(f,g)} = 0$ (complex bilinear dot product), so the parametrization is conformal, and holomorphicity implies the coordinate functions are harmonic; hence $\phi_{(f,g)}$ is minimal.

The induced metric (first fundamental form) of $\phi_{(f,g)}$ is conformal:

$$ds^2 = 2\langle \phi_z, \phi_{\bar{z}}\rangle\,|dz|^2 = \frac{|f|^2(1+|g|^2)^2}{4}\,|dz|^2.$$

In particular, the metric depends only on $|f|$ and $|g|$. Thus multiplying f by a unimodular constant $e^{-i\theta}$ leaves ds^2 unchanged.

Non-uniqueness arises from holomorphic reparametrization. If $\alpha : W \to D$ is biholomorphic and we reparametrize by $z = \alpha(w)$, then $dz = \alpha'(w)\,dw$ and $\phi_{(f,g)} \circ \alpha$ is again a Weierstrass immersion with new data

$$\tilde{g}(w) = g(\alpha(w)), \qquad \tilde{f}(w) = f(\alpha(w))\,\alpha'(w).$$

At points where $g'(z_0) \neq 0$, the holomorphic inverse function theorem gives a local inverse g^{-1}. Choosing $\alpha = g^{-1}$ yields $\tilde{g} = \mathrm{id}$ and $\tilde{f} =: F$ holomorphic, so locally one may normalize the data to (F, id) and write the immersion as ϕ_F. The family $\phi_{e^{-i\theta}F}$ $(\theta \in \mathbb{R})$ is called the associated family; since $|e^{-i\theta}F| = |F|$, all members have the same induced metric and are therefore locally isometric.

Geometric interpretation

The Weierstrass data (f, g) encode two geometric ingredients.

(1) The Gauss map. For a conformal minimal immersion $X = \phi_{(f,g)}$, let N be the unit normal field. The Gauss map is $N : D \to \mathbb{S}^2$. Composing with stereographic projection $\sigma : \mathbb{S}^2 \setminus \{(0,0,1)\} \to \mathbb{C}$ gives a (locally) meromorphic function

$$g = \sigma \circ N.$$

Thus $g(z)$ is the normal direction written in a complex coordinate. In particular, constant g means constant normal, hence X is a plane; nonconstant g corresponds to bending of the surface.

(2) The holomorphic 1-form $f\,dz$. The differential $f(z)\,dz$ controls the conformal scale of the parametrization. The induced metric is

$$ds^2 = \frac{|f|^2(1+|g|^2)^2}{4}\,|dz|^2,$$

so $|f|$ (together with $|g|$) determines local lengths on the surface, while the argument of f represents a constant "phase" in the complex differential.

(3) Conformality and minimality. The $\mathbb{C}^3$-valued 1-form

$$\Phi_{(f,g)}\,dz = \Big(\tfrac{1}{2}f(1-g^2),\ \tfrac{i}{2}f(1+g^2),\ fg\Big)\,dz$$

is arranged so that the complex derivative satisfies $X_z \cdot X_z = 0$, which is the isotropy condition ensuring the parametrization is conformal. Moreover each coordinate function of $X = \Re \int \Phi$ is harmonic (real part of a holomorphic function), hence X is minimal.

(4) Associated family. Multiplying f by a unimodular constant $e^{-i\theta}$ rotates the holomorphic differential by a constant phase. This generally changes the immersion in $\mathbb{R}^3$, but does not change the metric since $|e^{-i\theta}f| = |f|$. Therefore $\phi_{e^{-i\theta}F}$ all share the same first fundamental form and are locally isometric.

Theory (definitions and detailed metric computation)

Complex coordinate and derivatives. Let $z = x + iy$ on a domain $D \subset \mathbb{C}$. Then $|dz|^2 := dx^2 + dy^2$. We use $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$ and $\partial_{\bar z} = \frac{1}{2}(\partial_x + i\partial_y)$.

Weierstrass 1-form and immersion. Given holomorphic data (f, g) (with g meromorphic in general), define the $\mathbb{C}^3$-valued holomorphic 1-form

$$\Phi_{(f,g)}(z)\,dz = \Big(\tfrac{1}{2}f(1-g^2),\ \tfrac{i}{2}f(1+g^2),\ fg\Big)\,dz.$$

On simply connected D, set $\Psi(z) := \int^z \Phi_{(f,g)}(\zeta)\,d\zeta \in \mathbb{C}^3$ and define the immersion

$$\phi_{(f,g)}(z) = \Re\Psi(z) \in \mathbb{R}^3.$$

We distinguish the *complex bilinear* dot product $a \cdot b = \sum_{j=1}^3 a_j b_j$ from the Hermitian product $\langle a, b\rangle = \sum_{j=1}^3 a_j \overline{b_j}$.

Isotropy $\Phi \cdot \Phi = 0$ (conformality). Write

$$\Phi_1 = \tfrac{1}{2}f(1-g^2), \qquad \Phi_2 = \tfrac{i}{2}f(1+g^2), \qquad \Phi_3 = fg.$$

Then, using the complex bilinear dot product,

$$\Phi \cdot \Phi = \Phi_1^2 + \Phi_2^2 + \Phi_3^2 = \frac{f^2}{4}(1-g^2)^2 - \frac{f^2}{4}(1+g^2)^2 + f^2 g^2.$$

Since $(1-g^2)^2 - (1+g^2)^2 = -4g^2$, it follows that $\Phi \cdot \Phi = 0$. Moreover, Ψ is holomorphic with $\Psi'(z) = \Phi(z)$, hence

$$\phi_z = \partial_z \Re\Psi = \frac{1}{2}\Psi'(z) = \frac{1}{2}\Phi(z), \qquad \phi_{\bar z} = \frac{1}{2}\overline{\Phi(z)}.$$

Therefore $\phi_z \cdot \phi_z = \frac{1}{4}(\Phi \cdot \Phi) = 0$, which is the conformality condition.

Induced metric. In complex notation, the first fundamental form of a conformal immersion is

$$ds^2 = 2\langle \phi_z, \phi_{\bar z}\rangle \, |dz|^2.$$

Using $\phi_z = \frac{1}{2}\Phi$ we get

$$\langle \phi_z, \phi_{\bar z}\rangle = \left\langle \tfrac{1}{2}\Phi, \tfrac{1}{2}\overline{\Phi}\right\rangle = \frac{1}{4}\langle \Phi, \overline{\Phi}\rangle = \frac{1}{4}|\Phi|^2,$$

hence $ds^2 = \frac{|\Phi|^2}{2}|dz|^2$, where $|\Phi|^2 := |\Phi_1|^2 + |\Phi_2|^2 + |\Phi_3|^2$. Now

$$|\Phi_1|^2 = \frac{|f|^2}{4}|1-g^2|^2, \quad |\Phi_2|^2 = \frac{|f|^2}{4}|1+g^2|^2, \quad |\Phi_3|^2 = |f|^2|g|^2,$$

so

$$|\Phi|^2 = \frac{|f|^2}{4}\Big(|1-g^2|^2 + |1+g^2|^2 + 4|g|^2\Big).$$

Expanding gives

$$|1-g^2|^2 + |1+g^2|^2 = 2(1+|g|^4),$$

hence the bracket equals

$$2(1+|g|^4) + 4|g|^2 = 2(1+2|g|^2+|g|^4) = 2(1+|g|^2)^2.$$

Therefore

$$|\Phi|^2 = \frac{|f|^2}{2}(1+|g|^2)^2, \qquad ds^2 = \frac{|\Phi|^2}{2}|dz|^2 = \frac{|f|^2(1+|g|^2)^2}{4}\,|dz|^2.$$

Why minimal. Each component Ψ_k is holomorphic and $\phi_k = \Re\Psi_k$ is harmonic, so $\Delta\phi = 0$. For a conformal immersion, $\Delta\phi = 2\lambda^2 H$ where H is the mean curvature vector. Thus $\Delta\phi = 0$ implies $H = 0$, i.e. ϕ is minimal.

Lemma (metric in complex notation)

Let $\phi = \phi(x,y)$ be a smooth map into $\mathbb{R}^3$ and set

$$E = \langle \phi_x, \phi_x\rangle, \qquad F = \langle \phi_x, \phi_y\rangle, \qquad G = \langle \phi_y, \phi_y\rangle,$$

so that the first fundamental form is

$$ds^2 = E\,dx^2 + 2F\,dx\,dy + G\,dy^2.$$

Define complex derivatives

$$\phi_z = \tfrac{1}{2}(\phi_x - i\phi_y), \qquad \phi_{\bar z} = \tfrac{1}{2}(\phi_x + i\phi_y).$$

Then

$$\langle \phi_z, \phi_{\bar z}\rangle = \big\langle \tfrac{1}{2}(\phi_x - i\phi_y), \tfrac{1}{2}(\phi_x + i\phi_y)\big\rangle = \frac{1}{4}\big(\langle \phi_x, \phi_x\rangle + \langle \phi_y, \phi_y\rangle\big) = \frac{E+G}{4}.$$

In particular, if ϕ is conformal, then $F = 0$ and $E = G = \lambda^2$, hence

$$\langle \phi_z, \phi_{\bar z}\rangle = \frac{\lambda^2}{2}, \qquad \text{and therefore} \qquad ds^2 = \lambda^2(dx^2 + dy^2) = \lambda^2 |dz|^2 = 2\langle \phi_z, \phi_{\bar z}\rangle\, |dz|^2.$$

Companion identity (conformality test)

With $\phi_z = \frac{1}{2}(\phi_x - i\phi_y)$ we compute

$$\langle \phi_z, \phi_z\rangle = \big\langle \tfrac{1}{2}(\phi_x - i\phi_y), \tfrac{1}{2}(\phi_x - i\phi_y)\big\rangle = \frac{1}{4}\Big(\langle \phi_x, \phi_x\rangle - \langle \phi_y, \phi_y\rangle - 2i\langle \phi_x, \phi_y\rangle\Big) = \frac{E - G - 2iF}{4}.$$

Hence $\langle \phi_z, \phi_z\rangle = 0$ if and only if $E = G$ and $F = 0$, i.e. if and only if the parametrization is conformal.

Step-by-step origin of the Weierstrass data (f, g)

We want a systematic way to write all conformal minimal immersions $\phi : D \subset \mathbb{C} \to \mathbb{R}^3$ using holomorphic data. The Weierstrass representation gives exactly that: two complex functions (f, g) generate the surface.

1) Start from a minimal immersion and use complex coordinates.
Let $z = x + iy$ be a complex parameter on D. For a smooth map $\phi(x, y) \in \mathbb{R}^3$, define

$$\phi_z = \frac{1}{2}(\phi_x - i\phi_y), \qquad \phi_{\bar z} = \frac{1}{2}(\phi_x + i\phi_y).$$

A parametrization is **conformal** if and only if

$$\langle \phi_x, \phi_x\rangle = \langle \phi_y, \phi_y\rangle, \qquad \langle \phi_x, \phi_y\rangle = 0.$$

Equivalently, in complex form,

$$\phi_z \cdot \phi_z = 0 \qquad \text{(complex bilinear dot product).}$$

A surface is **minimal** if its coordinate functions are harmonic in conformal parameters, i.e.

$$\Delta\phi = 0.$$

In complex notation, $\Delta = 4\,\partial_z \partial_{\bar z}$, so $\Delta\phi = 0$ means

$$\partial_{\bar z}\phi_z = 0,$$

i.e. ϕ_z is holomorphic as a $\mathbb{C}^3$-valued function.

So:

$$\boxed{\text{Minimal + conformal} \implies \phi_z \text{ is a holomorphic } \mathbb{C}^3\text{-valued function satisfying } \phi_z \cdot \phi_z = 0.}$$

2) The holomorphic 1-form. Set

$$\Phi(z) := 2\phi_z(z) \in \mathbb{C}^3.$$

Then Φ is holomorphic and satisfies $\Phi \cdot \Phi = 0$. Since D is simply connected, the integral

$$\Psi(z) = \int^z \Phi(\zeta)\, d\zeta \in \mathbb{C}^3$$

is well-defined (up to an additive constant), and the immersion is recovered by

$$\phi(z) = \Re\Psi(z) = \Re \int^z \Phi(\zeta)\, d\zeta.$$

3) Introducing (f, g) from $\Phi \cdot \Phi = 0$. Write $\Phi = (\Phi_1, \Phi_2, \Phi_3)$ with $\Phi_1^2 + \Phi_2^2 + \Phi_3^2 = 0$. On the set where $\Phi_1 - i\Phi_2 \neq 0$, define

$$f := \Phi_1 - i\Phi_2, \qquad g := \frac{\Phi_3}{\Phi_1 - i\Phi_2} = \frac{\Phi_3}{f}.$$

Then $\Phi_3 = fg$ and

$$(\Phi_1 - i\Phi_2)(\Phi_1 + i\Phi_2) + \Phi_3^2 = 0 \quad \implies \quad f(\Phi_1 + i\Phi_2) + (fg)^2 = 0 \quad \implies \quad \Phi_1 + i\Phi_2 = -fg^2.$$

Hence

$$\begin{aligned} \Phi_1 &= \tfrac{1}{2}\big((\Phi_1 - i\Phi_2) + (\Phi_1 + i\Phi_2)\big) = \tfrac{1}{2} f(1 - g^2), \\ \Phi_2 &= \tfrac{1}{2i}\big((\Phi_1 + i\Phi_2) - (\Phi_1 - i\Phi_2)\big) = \tfrac{i}{2} f(1 + g^2), \\ \Phi_3 &= fg. \end{aligned}$$

Therefore

$$\Phi_{(f,g)}(z)\, dz = \Big(\tfrac{1}{2} f(1 - g^2),\ \tfrac{i}{2} f(1 + g^2),\ fg\Big)\, dz, \qquad \phi_{(f,g)}(z) = \Re \int^z \Phi_{(f,g)}(\zeta)\, d\zeta.$$

Geometrically, g is the stereographic coordinate of the Gauss map (normal direction), and $f\, dz$ is the holomorphic differential controlling the conformal scale.

4) Metric and associated family. The induced metric is conformal and equals

$$ds^2 = \frac{|f|^2(1 + |g|^2)^2}{4}\, |dz|^2.$$

In particular ds^2 depends only on $|f|$ and $|g|$, so replacing f by $e^{-i\theta} f$ leaves ds^2 unchanged.

5) Reparametrization. If $\alpha : W \to D$ is biholomorphic and $z = \alpha(w)$, then $dz = \alpha'(w)\, dw$ and the pullback gives new Weierstrass data

$$\tilde{g}(w) = g(\alpha(w)), \qquad \tilde{f}(w) = f(\alpha(w))\, \alpha'(w).$$

What are f and g? How $\mathbb{C}^3$ appears, and why the surface is in $\mathbb{R}^3$.

Complex derivative of a conformal minimal immersion. Start with a (conformal) minimal immersion

$$\phi : D \subset \mathbb{C} \longrightarrow \mathbb{R}^3.$$

Write $z = x + iy$ and define the complex derivatives

$$\phi_z = \frac{1}{2}(\phi_x - i\phi_y) \in \mathbb{C}^3, \qquad \phi_{\bar{z}} = \frac{1}{2}(\phi_x + i\phi_y) \in \mathbb{C}^3.$$

For a conformal minimal immersion, ϕ_z has two key properties:

- **Holomorphic:** minimality implies $\Delta\phi = 0$, and since $\Delta = 4\,\partial_z\partial_{\bar{z}}$ we obtain

  $$\partial_{\bar{z}}\phi_z = 0,$$

  so ϕ_z is holomorphic as a $\mathbb{C}^3$-valued function.

- **Isotropic:** conformality is equivalent to

  $$\phi_z \cdot \phi_z = 0 \qquad \text{(complex bilinear dot product).}$$

The holomorphic $\mathbb{C}^3$-valued 1-form. Now set

$$\Phi := 2\phi_z \in \mathbb{C}^3.$$

Then Φ is a holomorphic map into $\mathbb{C}^3$ satisfying $\Phi \cdot \Phi = 0$. The 1-form $\Phi\,dz$ is the holomorphic $\mathbb{C}^3$-valued form that one integrates to recover ϕ:

$$\Psi(z) = \int^z \Phi(\zeta)\,d\zeta \in \mathbb{C}^3, \qquad \phi(z) = \Re\Psi(z) \in \mathbb{R}^3.$$

Geometric meaning. The function g is (up to stereographic projection) the Gauss map: it encodes the unit normal direction $N \in \mathbb{S}^2$. Concretely,

$$N = \frac{1}{1+|g|^2}\big(2\Re g,\ 2\Im g,\ |g|^2 - 1\big),$$

which is inverse stereographic projection from $g \in \mathbb{C}$ to $N \in \mathbb{S}^2$. The holomorphic 1-form $f\,dz$ controls the conformal scale; the induced metric is

$$ds^2 = \frac{|f|^2(1+|g|^2)^2}{4}\,|dz|^2.$$

Thus $|f|$ (together with $|g|$) determines local lengths; multiplying f by $e^{-i\theta}$ rotates the differential but does not change the metric.

Clarifying the target spaces ($\mathbb{C}^3$ vs. $\mathbb{R}^3$ vs. $\mathbb{C}^2$).

1. The surface itself is in $\mathbb{R}^3$: $\phi(D) \subset \mathbb{R}^3$.

2. The space $\mathbb{C}^3$ appears because the complex derivative ϕ_z is naturally complex-valued, so $\phi_z \in \mathbb{C}^3$ and hence $\Phi = 2\phi_z \in \mathbb{C}^3$. We integrate in $\mathbb{C}^3$ and then take the real part:

   $$\Psi(z) = \int^z \Phi(\zeta)\,d\zeta \in \mathbb{C}^3, \qquad \phi(z) = \Re\Psi(z) \in \mathbb{R}^3.$$

3. The data (f, g) are two complex functions, hence define a map $(f, g) : D \to \mathbb{C}^2$, but (f, g) is *not* the surface. It is the holomorphic data used to build Φ and then ϕ.

One-sentence summary. The surface is $\phi : D \subset \mathbb{C} \to \mathbb{R}^3$; the complex derivative produces $\Phi = 2\phi_z \in \mathbb{C}^3$ with $\Phi \cdot \Phi = 0$; and the functions $f = \Phi_1 - i\Phi_2$ and $g = \Phi_3/f$ encode Φ, with g representing the Gauss map (stereographic coordinate) and $f\,dz$ controlling the conformal scale.

Where do ϕ_x and ϕ_y come from?

A parametrized surface patch is a smooth map

$$\phi : D \subset \mathbb{R}^2 \to \mathbb{R}^3, \qquad (x, y) \mapsto \phi(x, y).$$

Hence it has the usual partial derivatives with respect to the two real parameters:

$$\phi_x(x, y) = \frac{\partial \phi}{\partial x}(x, y), \qquad \phi_y(x, y) = \frac{\partial \phi}{\partial y}(x, y).$$

Writing $\phi = (\phi^1, \phi^2, \phi^3)$, these are

$$\phi_x = \big(\partial_x \phi^1, \partial_x \phi^2, \partial_x \phi^3\big) \in \mathbb{R}^3, \qquad \phi_y = \big(\partial_y \phi^1, \partial_y \phi^2, \partial_y \phi^3\big) \in \mathbb{R}^3.$$

If we identify (x, y) with the complex coordinate $z = x + iy$, we often write $\phi(z)$ instead of $\phi(x, y)$; this is only a change of notation. The complex derivatives are defined by the linear combinations

$$\phi_z = \tfrac{1}{2}(\phi_x - i\phi_y), \qquad \phi_{\bar z} = \tfrac{1}{2}(\phi_x + i\phi_y),$$

which package the two real tangent vectors ϕ_x, ϕ_y into complex form.

Why minimality implies $\Delta\phi = 0$ (in conformal parameters)

A surface is *minimal* if and only if its mean curvature satisfies $H \equiv 0$. For an immersion $\phi : D \subset \mathbb{R}^2 \to \mathbb{R}^3$ there is the fundamental identity

$$\Delta_g \phi = 2H\, N,$$

where Δ_g is the Laplace–Beltrami operator of the induced metric g and N is the unit normal. Thus $H = 0$ is equivalent to $\Delta_g \phi = 0$.

If (x, y) are *conformal* (isothermal) coordinates, then

$$g = \lambda^2(dx^2 + dy^2) \qquad (E = G = \lambda^2,\ F = 0),$$

and the Laplace–Beltrami operator simplifies to

$$\Delta_g = \frac{1}{\lambda^2}\Delta, \qquad \Delta = \partial_x^2 + \partial_y^2.$$

Substituting into $\Delta_g \phi = 2HN$ yields

$$\frac{1}{\lambda^2}\Delta\phi = 2HN \quad \Longrightarrow \quad \Delta\phi = 2\lambda^2 HN.$$

Hence, for a minimal surface ($H = 0$), we obtain $\Delta\phi = 0$.

Finally, since $\Delta = 4\,\partial_z \partial_{\bar z}$, the equation $\Delta\phi = 0$ implies

$$\partial_{\bar z}\phi_z = \partial_{\bar z}(\partial_z \phi) = 0,$$

so ϕ_z is holomorphic as a $\mathbb{C}^3$-valued function.

Why ϕ_z is $\mathbb{C}^3$-valued although ϕ is $\mathbb{R}^3$-valued

Write the immersion as

$$\phi(x, y) = (\phi^1(x, y), \phi^2(x, y), \phi^3(x, y)) \in \mathbb{R}^3.$$

Then

$$\phi_x = (\partial_x\phi^1, \partial_x\phi^2, \partial_x\phi^3) \in \mathbb{R}^3, \qquad \phi_y = (\partial_y\phi^1, \partial_y\phi^2, \partial_y\phi^3) \in \mathbb{R}^3.$$

By definition,

$$\phi_z = \tfrac{1}{2}(\phi_x - i\phi_y),$$

so each component is a complex number:

$$\phi_z = \left(\tfrac{1}{2}(\partial_x\phi^1 - i\partial_y\phi^1),\ \tfrac{1}{2}(\partial_x\phi^2 - i\partial_y\phi^2),\ \tfrac{1}{2}(\partial_x\phi^3 - i\partial_y\phi^3)\right) \in \mathbb{C}^3.$$

Thus ϕ takes values in $\mathbb{R}^3$, but its complex derivative ϕ_z is a complex linear combination of real tangent vectors and naturally lives in the complexified space $\mathbb{C}^3$.

Moreover, saying that ϕ_z is *holomorphic as a $\mathbb{C}^3$-valued function* means

$$\partial_{\bar{z}}\phi_z = 0,$$

equivalently $\partial_{\bar{z}}(\phi^k_z) = 0$ for each component $k = 1, 2, 3$.

Returning to $\mathbb{R}^3$ after working in $\mathbb{C}^3$

Although $\phi : D \to \mathbb{R}^3$ is real-valued, its complex derivative

$$\phi_z = \tfrac{1}{2}(\phi_x - i\phi_y)$$

is $\mathbb{C}^3$-valued. Set

$$\Phi := 2\phi_z \in \mathbb{C}^3.$$

Since Φ is holomorphic, we can integrate componentwise to obtain a holomorphic map

$$\Psi(z) = \int^z \Phi(\zeta)\, d\zeta \in \mathbb{C}^3$$

(well-defined on simply connected domains, up to an additive constant). The original immersion is then recovered by taking the real part componentwise:

$$\phi(z) = \Re\Psi(z) = \Re \int^z \Phi(\zeta)\, d\zeta \in \mathbb{R}^3.$$

Thus we *temporarily* work in the complexified space $\mathbb{C}^3$ to exploit holomorphicity, and then take $\Re$ to return to a surface in $\mathbb{R}^3$.

Why g is the Gauss map, but f is not.

Let $X = \phi_{(f,g)} : D \subset \mathbb{C} \to \mathbb{R}^3$ be a conformal minimal immersion given by the Weierstrass data (f, g):

$$X(z) = \Re \int^z \Phi_{(f,g)}(\zeta)\, d\zeta, \qquad \Phi_{(f,g)}(z) = \left(\tfrac{1}{2}f(1 - g^2),\ \tfrac{i}{2}f(1 + g^2),\ fg\right).$$

The induced metric is
$$ds^2 = \frac{|f|^2(1+|g|^2)^2}{4}\,|dz|^2.$$

1) The Gauss map and the role of g. The *Gauss map* of X is the unit normal field
$$N : D \to \mathbb{S}^2.$$
Let $\sigma : \mathbb{S}^2 \setminus \{(0,0,1)\} \to \mathbb{C}$ be stereographic projection. A fundamental fact of the Weierstrass representation is that
$$g = \sigma \circ N,$$
so g is (locally) the complex coordinate of the normal direction. Equivalently, N can be recovered from g by inverse stereographic projection:
$$N = \frac{1}{1+|g|^2}\big(2\Re g,\ 2\Im g,\ |g|^2-1\big).$$
Thus g *encodes the direction of the unit normal*, i.e. the Gauss map.

2) The role of f (conformal scale, not normals). The function f is *not* a Gauss map. It controls the conformal scale of the parametrization via the metric:
$$ds^2 = \frac{|f|^2(1+|g|^2)^2}{4}\,|dz|^2.$$
In particular, $|f|$ determines local lengths in the parameter domain (together with $|g|$), while the argument of f is a constant "phase" in the holomorphic differential. There is no formula expressing N purely in terms of f.

3) Why it may look like "both are Gauss maps" after a reparametrization. Non-uniqueness comes from biholomorphic reparametrization. If $\alpha : W \to D$ is biholomorphic, then $X \circ \alpha$ is again a Weierstrass immersion with new data
$$\tilde g(w) = g(\alpha(w)), \qquad \tilde f(w) = f(\alpha(w))\,\alpha'(w).$$
If $g'(z_0) \neq 0$, then g has a local holomorphic inverse g^{-1} near z_0. Choosing $\alpha = g^{-1}$ gives
$$\tilde g(w) = g(g^{-1}(w)) = w = \mathrm{id}(w), \qquad \tilde f(w) = f(g^{-1}(w))\,(g^{-1})'(w) =: F(w).$$
So locally we may normalize the data to (F, id). This does *not* mean that F is a Gauss map; it only means we chose coordinates so that the Gauss-map coordinate becomes the parameter w itself.

Summary.

$$\boxed{g \text{ is (stereographic) Gauss map data, while } f\,dz \text{ controls the conformal scale of the immersion.}}$$

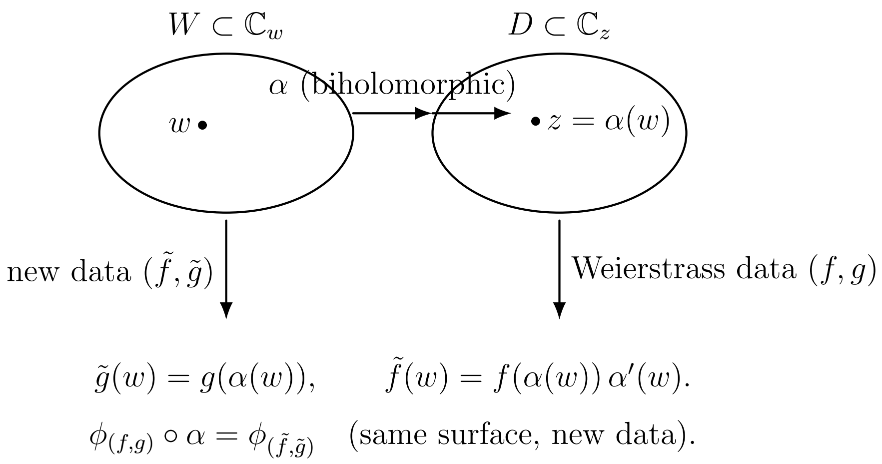


Figure 9: Holomorphic reparametrization: pull back the Weierstrass data by $z = \alpha(w)$.

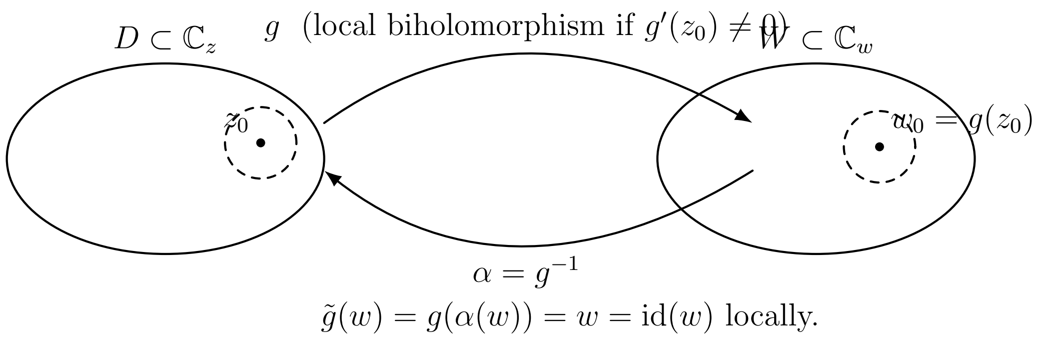


Figure 10: Local normalization near a regular point: if $g'(z_0) \neq 0$, choose $\alpha = g^{-1}$ so that $\tilde{g} = \mathrm{id}$.

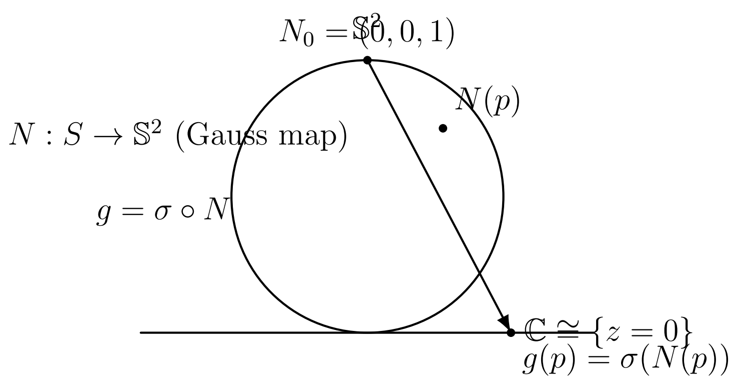


Figure 11: Geometric meaning of g: it is the Gauss map written in complex coordinates via stereographic projection.

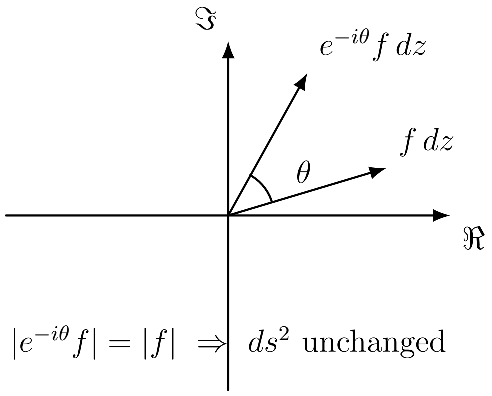


Figure 12: Associated family: multiply f by $e^{-i\theta}$ (constant phase). The metric depends on $|f|$, hence is unchanged.

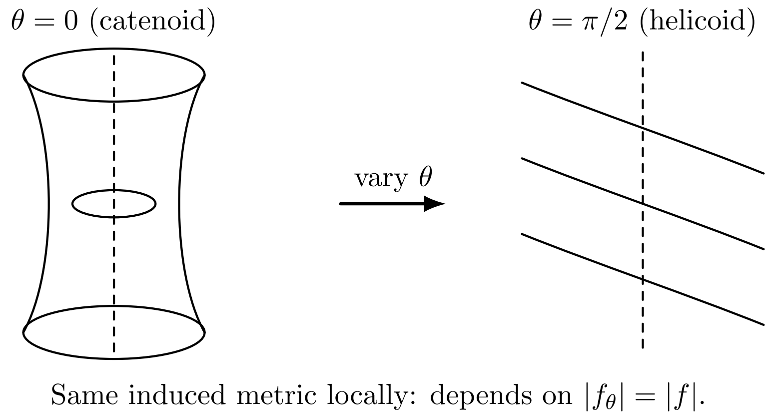


Figure 13: Classical picture: the associated family interpolates between catenoid and helicoid while preserving the local metric.

Idea (one picture). The Weierstrass pair (f, g) is not unique because we may reparametrize the domain by any biholomorphic map $\alpha : W \to D$ (change of complex coordinate). Under $z = \alpha(w)$, the same minimal immersion can be written again in Weierstrass form with new data

$$\tilde{g}(w) = g(\alpha(w)), \qquad \tilde{f}(w) = f(\alpha(w))\,\alpha'(w).$$

At a regular point z_0 of g (where $g'(z_0) \neq 0$), the holomorphic inverse function theorem allows us to choose $\alpha = g^{-1}$ locally, which normalizes the Gauss-map coordinate to $\tilde{g} = \mathrm{id}$ and leaves only one holomorphic function $F := \tilde{f}$. In this normalized form (F, id), multiplying F by a unimodular constant $e^{-i\theta}$ rotates the complex differential by a constant phase but does not change its modulus; hence the induced metric

$$ds^2 = \frac{|F|^2(1 + |z|^2)^2}{4}\,|dz|^2$$

is unchanged. Therefore the associated family $\{\phi_{e^{-i\theta}F}\}_{\theta\in\mathbb{R}}$ consists of locally isometric minimal surfaces.

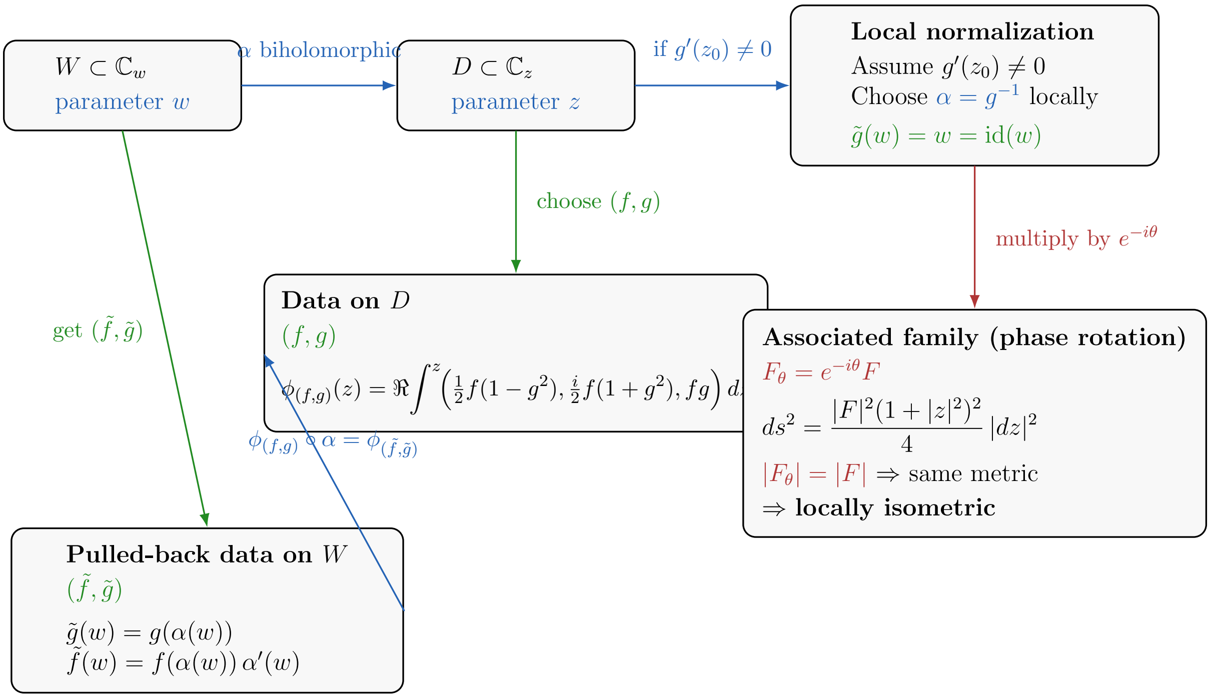


Figure 14: Problem 10 overview: holomorphic reparametrization changes (f, g), local inversion normalizes $g = \mathrm{id}$, and phase rotation $F \mapsto e^{-i\theta}F$ produces the associated family with the same induced metric.

Geometric meaning of g (Gauss map in complex coordinates). For a conformal minimal immersion $\phi : D \to \mathbb{R}^3$, let $N : D \to \mathbb{S}^2$ be the unit normal (Gauss map). Stereographic projection $\sigma : \mathbb{S}^2 \setminus \{N_0\} \to \mathbb{C}$ (from the north pole $N_0 = (0, 0, 1)$) identifies $\mathbb{S}^2$ (minus one point) with the complex plane. The Weierstrass function g is precisely this normal direction written as a complex number:

$$g = \sigma \circ N.$$

Thus g encodes how the surface bends (how the normal changes), while $f\, dz$ controls the conformal scale; the induced metric is

$$ds^2 = \frac{|f|^2(1 + |g|^2)^2}{4} |dz|^2,$$

so the modulus $|f|$ matters for lengths, whereas a constant phase factor $e^{-i\theta}$ does not change ds^2.

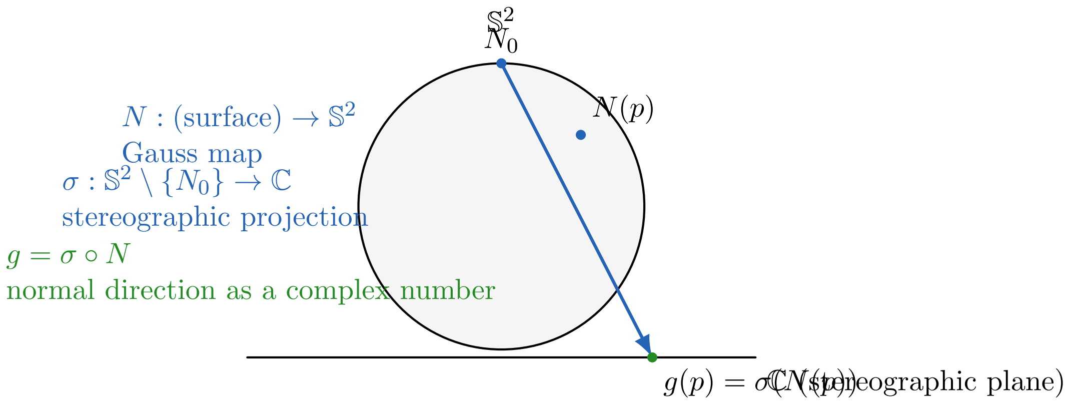


Figure 15: The Weierstrass function g is the Gauss map written in complex coordinates: $g = \sigma \circ N$.

Bidirectional view: g encodes the unit normal. Stereographic projection $\sigma : \mathbb{S}^2 \setminus \{N_0\} \to \mathbb{C}$ identifies (almost all of) the unit sphere with the complex plane. For a conformal immersion ϕ, the Gauss map $N : D \to \mathbb{S}^2$ gives the unit normal, and

$$g = \sigma \circ N$$

is exactly that normal direction written as a complex number. Conversely, g determines the unit normal by inverse stereographic projection:

$$N = \sigma^{-1}(g) = \frac{1}{1+|g|^2}\big(2\Re g,\ 2\Im g,\ |g|^2 - 1\big) \in \mathbb{S}^2.$$

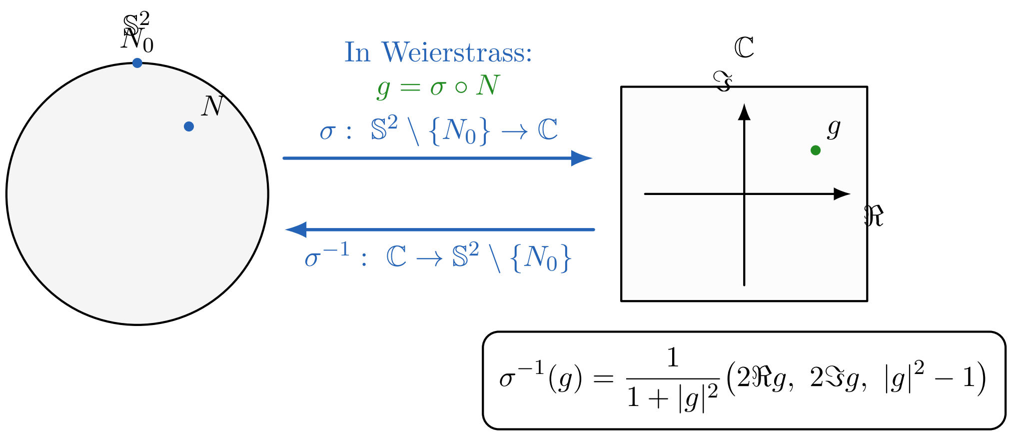


Figure 16: Stereographic projection and its inverse: a complex number g corresponds to a unit normal direction on $\mathbb{S}^2$.

Problem 11. Curvature, geodesics, and local isometry

Problem statement. Let S_1 and S_2 be surfaces with Gauss curvature K_{S_1}, K_{S_2} respectively.

1. Suppose $f : S_1 \to S_2$ is a diffeomorphism with $K_{S_2}(f(x)) = K_{S_1}(x)$. Must f be an isometry?

2. Suppose that for any geodesic $\gamma : I \to S_1$ parametrised by arc-length, $f \circ \gamma : I \to S_2$ is also a geodesic parametrised by arc-length. Must f be an isometry?

3. Let $p_1 \in S_1, p_2 \in S_2$. Suppose S_1 and S_2 are locally isometric around p_1 and p_2. Show that there exists a geodesic $\gamma_1(t)$ emanating from p_1 and $\gamma_2(t)$ emanating from p_2 such that $K_{S_1}(\gamma_1(t)) = K_{S_2}(\gamma_2(t))$, and such that if $\alpha_i(t)$ is a geodesic emanating from p_i and making an angle θ from γ_i (for fixed θ), then $K_{S_1}(\alpha_1(t)) = K_{S_2}(\alpha_2(t))$. Here all geodesics are parametrised by arc-length.

4. Show conversely that if the above property is satisfied, then the S_i are locally isometric near the p_i. Deduce that if S_1 and S_2 have constant curvature $K_{S_1} = K_{S_2}$, then S_1 and S_2 are locally isometric.

Hint: Consider geodesic polar coordinates and solve an ODE for G.

Solution.

1. **Not necessarily.** Take $S_1 = S_2 = \mathbb{S}^2$ (unit sphere). Then $K \equiv 1$, so $K_{S_2}(f(x)) = K_{S_1}(x)$ holds for every diffeomorphism $f : \mathbb{S}^2 \to \mathbb{S}^2$. Most diffeomorphisms are not isometries (they distort lengths/areas), hence the condition on curvature alone does not force f to be an isometry.

2. **Yes.** Fix $p \in S_1$ and let $v \in T_pS_1$ be a unit tangent vector. Let γ_v be the unit-speed geodesic with $\gamma_v(0) = p$ and $\gamma_v'(0) = v$. By hypothesis, $f \circ \gamma_v$ is also unit-speed, hence
$$\|df_p(v)\|_{S_2} = \|(f \circ \gamma_v)'(0)\|_{S_2} = 1.$$
Thus df_p maps the unit sphere in T_pS_1 into the unit sphere in $T_{f(p)}S_2$, so it preserves norms. By linearity and scaling, $\|df_p(w)\|_{S_2} = \|w\|_{S_1}$ for all $w \in T_pS_1$. Therefore for all $u, v \in T_pS_1$, polarization gives
$$\langle df_p(u), df_p(v)\rangle_{S_2} = \frac{\|df_p(u+v)\|^2 - \|df_p(u)\|^2 - \|df_p(v)\|^2}{2} = \frac{\|u+v\|^2 - \|u\|^2 - \|v\|^2}{2} = \langle u, v\rangle_{S_1}.$$
Hence df_p is an isometry on tangent spaces for every p, so f is a local isometry, and since it is a diffeomorphism, f is an isometry.

3. **Property (iii) from a local isometry.** Assume S_1 and S_2 are locally isometric near p_1, p_2. Let $F : U_1 \to U_2$ be a local isometry with $F(p_1) = p_2$. Choose any unit-speed geodesic γ_1 emanating from p_1 and define $\gamma_2 := F \circ \gamma_1$. Local isometries preserve arc-length and send geodesics to geodesics, so γ_2 is a unit-speed geodesic from p_2. Moreover Gauss curvature is intrinsic, so $K_{S_2}(F(x)) = K_{S_1}(x)$ on U_1, hence
$$K_{S_2}(\gamma_2(t)) = K_{S_1}(\gamma_1(t)).$$

If α_1 is any unit-speed geodesic from p_1 making angle θ with γ_1, define $\alpha_2 := F \circ \alpha_1$. Angles are preserved by a local isometry, so α_2 makes the same angle θ with γ_2, and again

$$K_{S_2}(\alpha_2(t)) = K_{S_1}(\alpha_1(t)).$$

4. **Conversely, curvature data in polar coordinates determines the metric.** Fix $p \in S$ and introduce geodesic polar coordinates (r, θ) around p (for r small). In these coordinates the metric takes the form

$$ds^2 = dr^2 + G(r,\theta)^2\, d\theta^2, \qquad G(0,\theta) = 0, \quad G_r(0,\theta) = 1.$$

A standard computation (using the Levi-Civita connection in these coordinates) yields the ODE

$$G_{rr}(r,\theta) + K(r,\theta)\, G(r,\theta) = 0,$$

where $K(r,\theta)$ is the Gauss curvature in the same polar coordinates.

Assume the property from (iii): for each fixed direction θ, the curvature functions along the corresponding radial geodesics agree between S_1 and S_2. Then the functions $G_1(r,\theta)$ and $G_2(r,\theta)$ satisfy the same ODE in r with the same initial conditions, hence

$$G_1(r,\theta) = G_2(r,\theta)$$

for r small. Therefore the metrics coincide in these polar coordinates, so identifying points with the same (r,θ) defines a local isometry near p_1 and p_2.

In particular, if $K_{S_1} = K_{S_2}$ is constant, then G is determined (locally) solely by this constant, so any two surfaces of the same constant Gauss curvature are locally isometric.

Examples.

1. **Flat case.** The plane and the cylinder both have $K \equiv 0$ and are locally isometric (unwrap the cylinder).
2. **Positive curvature.** Any surface with $K \equiv K_0 > 0$ is locally isometric to the sphere of radius $1/\sqrt{K_0}$.
3. **Negative curvature.** Any surface with constant $K_0 < 0$ is locally isometric to the hyperbolic plane of curvature K_0.

Problem 12. Intrinsic distance on a surface; Hopf–Rinow consequences

Problem statement. Let $S \subset \mathbb{R}^3$ be a smooth embedded surface. For $p, q \in S$ define the *intrinsic distance*

$$d(p,q) := \inf_{\alpha \in \Omega(p,q)} \ell(\alpha),$$

where $\Omega(p,q)$ is the set of piecewise C^1 curves $\alpha : [a,b] \to S$ with $\alpha(a) = p$, $\alpha(b) = q$, and $\ell(\alpha)$ is the length computed with the induced Riemannian metric on S.

1. Show that d is a metric on S, compatible with the topology of S. If S is complete (and without boundary), Hopf–Rinow implies that for any p, q there is a minimizing geodesic γ joining them with $d(p,q) = \ell(\gamma)$ and geodesics exist for all $t \in \mathbb{R}$.

(i) If $f : S_1 \to S_2$ is an isometry, show that $d_2(f(p), f(q)) = d_1(p, q)$ for all $p, q \in S_1$.

(ii) A geodesic $\gamma : [0, \infty) \to S$ is a *ray leaving from* p if $\gamma(0) = p$ and $d(\gamma(0), \gamma(s)) = \ell(\gamma|_{[0,s]}) = s$ for all $s \geq 0$. Let p be a point in a complete, noncompact surface S. Show that S contains a ray leaving from p. (You may assume geodesics depend smoothly, hence continuously, on their initial data.)

(iii) Let S be connected and let $p \in S$ be such that *all geodesics through* p *are closed*, i.e. each geodesic γ with $\gamma(0) = p$ extends to a smooth periodic map $\gamma : S^1 \to S$. Show that S is compact.

Solution.

(a) d **is a metric and induces the topology.**
Nonnegativity and symmetry. Lengths satisfy $\ell(\alpha) \geq 0$ and $\ell(\alpha) = \ell(\alpha^{-1})$ where $\alpha^{-1}(t) = \alpha(a + b - t)$, hence $d(p, q) \geq 0$ and $d(p, q) = d(q, p)$.

Triangle inequality. If $\alpha \in \Omega(p, q)$ and $\beta \in \Omega(q, r)$ then the concatenation $\alpha * \beta \in \Omega(p, r)$ satisfies $\ell(\alpha * \beta) = \ell(\alpha) + \ell(\beta)$, hence

$$d(p, r) \leq \ell(\alpha * \beta) = \ell(\alpha) + \ell(\beta).$$

Taking infima over α, β gives $d(p, r) \leq d(p, q) + d(q, r)$.

Definiteness. Clearly $d(p, p) = 0$. If $p \neq q$, choose a coordinate chart $\varphi : U \to \mathbb{R}^2$ with $p, q \in U$. Because the induced metric tensor is smooth and positive definite, on a sufficiently small compact neighborhood $K \subset U$ there exist constants $0 < c \leq C$ such that for all $x \in K$ and all $v \in \mathbb{R}^2$,

$$c\,|v| \leq \sqrt{g_x(v, v)} \leq C\,|v|.$$

Let α be any curve in K joining p to q. Writing $\tilde{\alpha} = \varphi \circ \alpha$,

$$\ell(\alpha) = \int \sqrt{g_{\tilde{\alpha}(t)}(\tilde{\alpha}'(t), \tilde{\alpha}'(t))}\,dt \;\geq\; c \int |\tilde{\alpha}'(t)|\,dt \;\geq\; c\,|\tilde{p} - \tilde{q}|,$$

where $\tilde{p} = \varphi(p)$, $\tilde{q} = \varphi(q)$. Since $\tilde{p} \neq \tilde{q}$, this yields $\ell(\alpha) \geq c|\tilde{p} - \tilde{q}| > 0$, hence $d(p, q) > 0$. Therefore $d(p, q) = 0$ implies $p = q$.

Compatibility with the topology. Using the same local comparison, for p, q sufficiently close in a chart one has two-sided estimates

$$c\,|\varphi(p) - \varphi(q)| \;\leq\; d(p, q) \;\leq\; C\,|\varphi(p) - \varphi(q)|.$$

The right inequality follows by taking the curve to be the coordinate straight segment in $\mathbb{R}^2$ pulled back to S. These inequalities show that d-balls and Euclidean balls in coordinates contain each other locally, hence d induces the same open sets as the submanifold topology on S.

(i) Isometries preserve intrinsic distances. Let $f : S_1 \to S_2$ be an isometry. By definition, it preserves lengths of curves: for any piecewise C^1 curve α in S_1,

$$\ell_{S_2}(f \circ \alpha) = \ell_{S_1}(\alpha).$$

Therefore

$$d_2(f(p), f(q)) = \inf_{\beta \in \Omega(f(p), f(q))} \ell_{S_2}(\beta) \;\leq\; \inf_{\alpha \in \Omega(p,q)} \ell_{S_2}(f \circ \alpha) = \inf_{\alpha \in \Omega(p,q)} \ell_{S_1}(\alpha) = d_1(p, q).$$

Applying the same argument to the inverse isometry f^{-1} gives the reverse inequality, hence $d_2(f(p), f(q)) = d_1(p, q)$.

(ii) Existence of a ray from p on a complete noncompact surface. Because S is complete and noncompact, the distance balls $\overline{B}(p, R)$ cannot exhaust S with R bounded. Choose a sequence $q_n \in S$ such that

$$d(p, q_n) \xrightarrow[n\to\infty]{} \infty.$$

By Hopf–Rinow, for each n there exists a minimizing geodesic $\gamma_n : [0, L_n] \to S$ joining p to q_n, parametrized by arc length, with $L_n = d(p, q_n)$ and $\gamma_n(0) = p$.

Let $v_n = \gamma_n'(0) \in T_pS$; then $|v_n| = 1$. The unit circle in T_pS is compact, so passing to a subsequence we may assume $v_n \to v$ with $|v| = 1$. Let $\gamma : [0, \infty) \to S$ be the (unit-speed) geodesic with initial data $\gamma(0) = p$, $\gamma'(0) = v$. By the assumed continuous dependence on initial conditions, for each fixed $s > 0$,

$$\gamma_n(s) \to \gamma(s) \qquad \text{as } n \to \infty$$

for all large n (since $L_n \to \infty$ ensures $\gamma_n(s)$ is defined).

For each such n, the minimizing property of γ_n gives

$$d\big(p, \gamma_n(s)\big) = s.$$

Passing to the limit and using continuity of d with respect to the manifold topology (from part (a)) yields

$$d\big(p, \gamma(s)\big) = s \qquad \forall s \geq 0.$$

Similarly, for $0 \leq s \leq t$,

$$d\big(\gamma_n(s), \gamma_n(t)\big) = t - s$$

(since the restriction of a minimizing geodesic is minimizing). Passing to the limit gives

$$d\big(\gamma(s), \gamma(t)\big) = t - s \qquad \forall\, 0 \leq s \leq t.$$

Thus γ realizes distance on every subinterval and is a ray leaving from p.

(iii) If all geodesics through p are closed, then S is compact. For each unit vector $v \in T_pS$ let $\gamma_v : \mathbb{R} \to S$ be the unit-speed geodesic with $\gamma_v(0) = p$, $\gamma_v'(0) = v$. By hypothesis, γ_v is periodic: there exists $T(v) > 0$ such that $\gamma_v(t + T(v)) = \gamma_v(t)$ for all t. Using smooth dependence of γ_v on v, one may choose $T(v)$ continuously on the unit circle $S^1 \subset T_pS$. Since S^1 is compact, there is a finite maximum

$$T_{\max} := \max_{|v|=1} T(v) < \infty.$$

Fix v and consider any point $\gamma_v(t)$ on the closed geodesic. Since γ_v returns to p after time $T(v)$, the shorter of the two arcs from p to $\gamma_v(t)$ along the loop has length at most $T(v)/2$. Therefore

$$d\big(p, \gamma_v(t)\big) \leq \frac{T(v)}{2} \leq \frac{T_{\max}}{2}.$$

Hence every geodesic through p is contained in the closed ball $\overline{B}\big(p, \frac{T_{\max}}{2}\big)$.

Finally, in a connected surface, every point $q \in S$ can be joined to p by a minimizing geodesic segment (this is ensured, for instance, when S is complete; and the present hypothesis that *all* geodesics from p exist for all time implies geodesic completeness at p, which is the key input to apply the usual minimizing argument). Thus every $q \in S$ lies on some geodesic through p, and the estimate above gives

$$S \subset \overline{B}\big(p, \tfrac{T_{\max}}{2}\big)\,.$$

In a complete Riemannian surface, closed and bounded sets are compact (Hopf–Rinow), so the closed ball on the right is compact. Therefore S is a closed subset of a compact set, hence compact.

Examples.

1. **Plane.** For $S = \mathbb{R}^2$ with the standard metric, $d(p,q) = |p - q|$ and rays are straight half-lines.

2. **Sphere.** For $S = \mathbb{S}^2$, $d(p,q)$ is the great-circle distance. All geodesics are closed, and S is compact.

3. **Hyperbolic plane.** Complete and noncompact; from any p there exist many rays.

Problem 12. Intrinsic distance on a surface; Hopf–Rinow consequences

Problem statement. Let $S \subset \mathbb{R}^3$ be a smooth embedded surface with the induced Riemannian metric. For $p, q \in S$ define the intrinsic distance

$$d(p,q) = \inf_{\alpha \in \Omega(p,q)} \ell(\alpha),$$

where $\Omega(p,q)$ is the set of piecewise C^1 curves $\alpha : [a,b] \to S$ with $\alpha(a) = p$, $\alpha(b) = q$, and $\ell(\alpha)$ is the (Riemannian) length of α. Show that d is a metric compatible with the topology of S. If S is complete (and without boundary), Hopf–Rinow implies that for any $p, q \in S$ there exists a minimizing geodesic γ with $d(p,q) = \ell(\gamma)$ and γ defined for all $t \in \mathbb{R}$.

(i) If $f : S_1 \to S_2$ is an isometry, show $d_2(f(p), f(q)) = d_1(p,q)$.

(ii) A geodesic $\gamma : [0,\infty) \to S$ is a *ray from p* if $\gamma(0) = p$ and $d(\gamma(0), \gamma(s)) = s$ for all $s \geq 0$. If S is complete and noncompact, show S contains a ray from any p.

(iii) If S is connected and there exists $p \in S$ such that *all geodesics through p are closed*, show that S is compact.

Theory / setup.

1) Induced Riemannian metric and curve length. The Euclidean inner product on $\mathbb{R}^3$ induces a Riemannian metric on S by restriction to tangent planes: for $v, w \in T_xS$,

$$g_x(v,w) = \langle v, w \rangle_{\mathbb{R}^3}.$$

If $\alpha : [a,b] \to S$ is piecewise C^1, its length is

$$\ell(\alpha) = \int_a^b \|\alpha'(t)\|\,dt, \qquad \|\alpha'(t)\| := \sqrt{g_{\alpha(t)}(\alpha'(t), \alpha'(t))}.$$

In a local parametrization $X : U \subset \mathbb{R}^2 \to S$ with coordinates (u^1, u^2), the metric has coefficients $g_{ij} = \langle X_{u^i}, X_{u^j} \rangle$, and

$$\|\alpha'(t)\|^2 = g_{ij}(u(t))\, \dot{u}^i(t)\, \dot{u}^j(t).$$

2) Intrinsic distance and why it is a metric. Define

$$d(p, q) = \inf\{\ell(\alpha) : \alpha \in \Omega(p, q)\}.$$

Then:

- $d(p, q) \geq 0$ and $d(p, q) = d(q, p)$ because length is nonnegative and invariant under reversing the curve.
- Triangle inequality holds because concatenation satisfies $\ell(\alpha * \beta) = \ell(\alpha) + \ell(\beta)$.
- Definiteness ($d(p, q) = 0 \Rightarrow p = q$) follows from local comparison with Euclidean distance in a chart: since g_{ij} is continuous and positive definite, on a sufficiently small coordinate neighborhood there exist constants $0 < c \leq C$ such that

  $$c\,|v|^2 \leq g_{ij}(u)v^i v^j \leq C\,|v|^2,$$

  hence for any curve α inside that neighborhood,

  $$\ell(\alpha) \geq \sqrt{c}\,\ell_{\text{Euc}}(\alpha) \;\; \geq \;\; \sqrt{c}\,|u(p) - u(q)|.$$

  So if $p \neq q$ then $d(p, q) > 0$.

3) Compatibility with the manifold topology. The same local inequalities imply that, in a coordinate chart, intrinsic balls are comparable to Euclidean balls: for q close to p,

$$\sqrt{c}\,|u(p) - u(q)| \;\leq\; d(p, q) \;\leq\; \sqrt{C}\,|u(p) - u(q)|.$$

Thus the metric topology induced by d coincides with the usual surface topology.

4) Geodesics and minimizing geodesics. A smooth curve γ is a geodesic if $\nabla_{\dot{\gamma}}\dot{\gamma} = 0$ (Levi–Civita connection), i.e. locally it solves the geodesic ODE. If γ is parametrized by arc length, then $\|\dot{\gamma}\| \equiv 1$ and

$$\ell(\gamma|_{[s_0,s_1]}) = s_1 - s_0.$$

A *minimizing geodesic segment* from p to q is a geodesic $\gamma : [0, L] \to S$ with $\gamma(0) = p$, $\gamma(L) = q$ and

$$d(p, q) = \ell(\gamma) = L.$$

Geodesics are always *locally* minimizing for sufficiently small subintervals, but not necessarily globally minimizing.

5) Completeness notions. Let (S, d) be the metric space from intrinsic distance, and let (S, g) be the Riemannian surface.

- *Metric completeness*: every Cauchy sequence in (S, d) converges in S.
- *Geodesic completeness*: every geodesic γ extends to a solution defined for all $t \in \mathbb{R}$.

6) Hopf–Rinow theorem (what you use here). For a connected Riemannian manifold (M, g), the following are equivalent:

(HR1) (M, d) is complete as a metric space.

(HR2) Every closed and bounded set is compact; equivalently, closed metric balls $\overline{B}(p, R)$ are compact.

(HR3) Any two points $p, q \in M$ can be joined by a minimizing geodesic segment.

(HR4) M is geodesically complete (all geodesics exist for all $t \in \mathbb{R}$).

In Problem 12 one typically uses (HR1)⇒(HR3) and (HR1)⇒(HR2).

7) Isometries preserve intrinsic distance (tool for (i)). If $f : (S_1, g_1) \to (S_2, g_2)$ is an isometry, then for any piecewise C^1 curve α in S_1,

$$\ell_2(f \circ \alpha) = \ell_1(\alpha),$$

so taking infima over all curves joining p to q yields

$$d_2(f(p), f(q)) = d_1(p, q).$$

8) Rays and why completeness + noncompactness forces them (tool for (ii)). Fix $p \in S$. Noncompactness implies there exist points q_n with $d(p, q_n) \to \infty$. By Hopf–Rinow, for each n there is a minimizing unit-speed geodesic segment

$$\gamma_n : [0, L_n] \to S, \qquad L_n = d(p, q_n) \to \infty, \qquad \gamma_n(0) = p,\ \gamma_n(L_n) = q_n.$$

Choose a convergent subsequence of initial unit tangents $\gamma_n'(0) \in T_pS$. Using continuous dependence of geodesics on initial data, one obtains a limiting geodesic $\gamma : [0, \infty) \to S$ such that every initial segment $\gamma|_{[0,T]}$ is a limit of minimizing segments, hence remains minimizing:

$$d(p, \gamma(s)) = s \qquad \forall s \geq 0.$$

This γ is a *ray* from p.

9) Closed geodesics through one point force compactness (tool for (iii)). Assume every geodesic through p is closed. For each unit vector $v \in T_pS$ let γ_v be the unit-speed geodesic with $\gamma_v(0) = p$, $\gamma_v'(0) = v$, and let $T(v) > 0$ be a period (so $\gamma_v(T(v)) = p$). If one assumes (as in the problem hint) smooth/continuous dependence on initial data, one may choose $T(v)$ continuously in v (or at least show that the set of periods is locally bounded), and since the unit circle in T_pS is compact, obtain a uniform bound

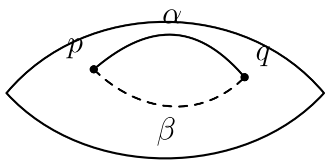
$$T(v) \leq T_{\max} < \infty \qquad \text{for all unit } v.$$

Then any point on any such closed geodesic is at intrinsic distance $\leq T_{\max}/2$ from p (go along the shorter arc of the loop). Hence the union of all geodesics from p is contained in the closed ball $\overline{B}(p, T_{\max}/2)$. Finally, by Hopf–Rinow, every point $q \in S$ is joined to p by a minimizing geodesic segment, so q lies on some geodesic through p. Thus

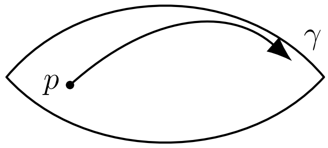
$$S \subset \overline{B}\left(p, \frac{T_{\max}}{2}\right),$$

and the right-hand side is compact (Hopf–Rinow), implying S is compact.

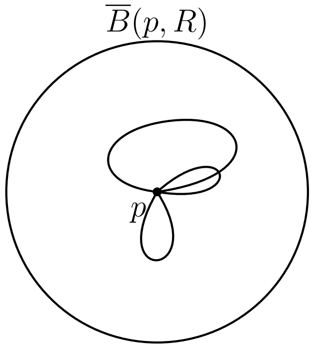
Schematic charts (optional). The following figures are schematic (topological cartoons), but are useful to remember what each part says.



$$d(p,q) = \inf\{\ell(\alpha) : \alpha \in \Omega(p,q)\}.$$



γ is a ray if $d(p, \gamma(s)) = s$ for all $s \geq 0$.



If all geodesics through p are closed and uniformly bounded, then $S \subset \overline{B}(p, R)$, hence compact.

Problem 13. Geodesics on the paraboloid $z = x^2 + y^2$ self-intersect infinitely often

Problem statement. Show that any geodesic of the paraboloid of revolution

$$z = x^2 + y^2$$

which is not a meridian intersects itself an infinite number of times. (*Hint: use Clairaut's relation. You may assume that no geodesic of a surface of revolution can be asymptotic to a parallel which is not itself a geodesic.*)

Solution.

1) Parametrization and metric. Use coordinates (r, θ), $r \geq 0$, $\theta \in \mathbb{R}/2\pi\mathbb{Z}$:

$$X(r, \theta) = (r \cos\theta,\ r \sin\theta,\ r^2).$$

Then

$$X_r = (\cos\theta,\ \sin\theta,\ 2r), \qquad X_\theta = (-r \sin\theta,\ r \cos\theta,\ 0),$$

so the first fundamental form is

$$ds^2 = E(r)\, dr^2 + G(r)\, d\theta^2, \qquad E(r) = \langle X_r, X_r \rangle = 1 + 4r^2, \quad G(r) = \langle X_\theta, X_\theta \rangle = r^2.$$

2) Clairaut's relation (conserved "angular momentum"). Let $\gamma(s) = (r(s), \theta(s))$ be a unit-speed geodesic (s is arc length). For any surface of revolution with metric $E(r)\,dr^2 + G(r)\,d\theta^2$, the coordinate θ is cyclic, so the quantity

$$G(r)\,\theta'(s) = \text{constant}$$

is conserved along geodesics. Here $G(r) = r^2$, hence

$$r(s)^2\,\theta'(s) = c$$

for some constant $c \in \mathbb{R}$.

If γ is a meridian, then $\theta \equiv \theta_0$ and thus $c = 0$. Conversely, a *non-meridian* geodesic has $c \neq 0$.

3) The radial equation and existence of a unique turning point. Unit speed means

$$1 = E(r)\,r'(s)^2 + r(s)^2\,\theta'(s)^2 = (1 + 4r^2)\,r'(s)^2 + \frac{c^2}{r^2}.$$

Therefore

$$r'(s)^2 = \frac{1 - \frac{c^2}{r^2}}{1 + 4r^2}.$$

In particular, $r(s) \geq |c|$ for all s, and $r'(s) = 0$ precisely at $r = |c|$. Thus a non-meridian geodesic reaches a minimal distance $r_{\min} = |c| > 0$ from the axis and then turns back outward. After reparametrization we may assume the turning point occurs at $s = 0$, so

$$r(0) = |c|, \qquad r(s) \text{ decreases on } (-\infty, 0] \text{ and increases on } [0, \infty).$$

4) The angular change is infinite (logarithmic divergence). From $r^2\theta' = c$ and the formula for r', we obtain

$$\frac{d\theta}{dr} = \frac{\theta'}{r'} = \pm\frac{c}{r^2}\,\frac{\sqrt{1 + 4r^2}}{\sqrt{1 - \frac{c^2}{r^2}}}.$$

Define for $R > |c|$ the (positive) function

$$\Theta(R) := \int_{|c|}^{R} \frac{|c|}{r^2}\,\frac{\sqrt{1 + 4r^2}}{\sqrt{1 - \frac{c^2}{r^2}}}\,dr.$$

Then along the two branches (incoming $s < 0$ and outgoing $s > 0$) one has, up to an additive constant $\theta(0)$,

$$\theta_{\text{out}}(R) = \theta(0) + \Theta(R), \qquad \theta_{\text{in}}(R) = \theta(0) - \Theta(R),$$

whenever $r(s) = R$ on the corresponding branch.

For large r, we have $\sqrt{1 + 4r^2} \sim 2r$ and $\sqrt{1 - \frac{c^2}{r^2}} \to 1$, so the integrand satisfies

$$\frac{|c|}{r^2}\,\frac{\sqrt{1 + 4r^2}}{\sqrt{1 - \frac{c^2}{r^2}}} \sim \frac{|c|}{r^2} \cdot 2r = \frac{2|c|}{r}.$$

Hence

$$\Theta(R) \sim 2|c| \int^R \frac{dr}{r} = 2|c| \log R \xrightarrow[R\to\infty]{} \infty.$$

So as $R \to \infty$, both branches wind around the axis through arbitrarily large angles.

5) Infinitely many self-intersections. Consider points on the incoming and outgoing branches with the *same* radius R: they have polar angles differing by

$$\theta_{\text{out}}(R) - \theta_{\text{in}}(R) = 2\Theta(R).$$

Since $\Theta(R) \to \infty$ continuously with R, for every integer $k \geq 1$ there exists R_k such that

$$2\Theta(R_k) = 2\pi k.$$

At this radius, the incoming and outgoing points satisfy

$$r_{\text{out}} = r_{\text{in}} = R_k, \qquad \theta_{\text{out}} = \theta_{\text{in}} + 2\pi k,$$

hence they represent the *same point* on the surface. This gives a self-intersection for each k. Therefore the non-meridian geodesic intersects itself infinitely many times.

Examples.

1. **Meridians.** $\theta \equiv \theta_0$ are geodesics with no self-intersections (they are excluded by the statement).
2. **A typical non-meridian.** Fix $c \neq 0$. Then $r_{\min} = |c|$ and the geodesic spirals in to $r_{\min}$ and spirals back out, while accumulating unbounded angle; the two spiral branches cross each other infinitely often, producing infinitely many self-intersections.

Problem 13. Geodesics on the paraboloid $z = x^2+y^2$ and Clairaut's relation

Surface of revolution and the induced metric. Parametrize the paraboloid by polar coordinates:

$$X(r,\theta) = (r\cos\theta,\ r\sin\theta,\ r^2), \qquad r > 0,\ \theta \in \mathbb{R}.$$

Compute the tangent vectors:

$$X_r = (\cos\theta,\ \sin\theta,\ 2r), \qquad X_\theta = (-r\sin\theta,\ r\cos\theta,\ 0).$$

Hence the first fundamental form coefficients are

$$E = \langle X_r, X_r\rangle = 1 + 4r^2, \qquad F = \langle X_r, X_\theta\rangle = 0, \qquad G = \langle X_\theta, X_\theta\rangle = r^2.$$

Therefore the induced metric ("ds^2", i.e. the squared infinitesimal distance) is

$$ds^2 = (1 + 4r^2)\, dr^2 + r^2\, d\theta^2.$$

Geodesic Lagrangian and Clairaut's relation. For a curve $\gamma(s) = (r(s), \theta(s))$ on the surface, its speed squared is

$$|\gamma'(s)|^2 = (1 + 4r^2)\, (r')^2 + r^2(\theta')^2,$$

so we may take the energy Lagrangian

$$L(r, \theta, r', \theta') = \frac{1}{2}\Big((1 + 4r^2)(r')^2 + r^2(\theta')^2\Big).$$

Since L does not depend on θ (the surface is rotationally symmetric), the Euler–Lagrange equation for θ gives a conserved quantity (“angular momentum”):

$$\frac{d}{ds}\Big(\frac{\partial L}{\partial \theta'}\Big) = 0 \quad \Longrightarrow \quad \frac{\partial L}{\partial \theta'} = r^2\theta' = \text{const} =: c.$$

This is Clairaut’s relation in (r, θ)-form.

Geometric form of Clairaut. Let $\psi(s)$ be the angle between the unit-speed geodesic γ and the meridian direction X_r (i.e. $\psi = 0$ means “moving along a meridian”). Then one has the equivalent form

$$r(s)\,\sin\psi(s) = \text{const}.$$

Indeed, for unit speed,

$$1 = (1 + 4r^2)(r')^2 + r^2(\theta')^2,$$

and the component of velocity in the parallel direction is

$$\sin\psi = \frac{\text{(speed along parallels)}}{\text{total speed}} = \frac{\sqrt{G}\,|\theta'|}{1} = r|\theta'|.$$

Thus $r\sin\psi = r(r|\theta'|) = r^2|\theta'|$, so constancy of $r^2\theta'$ is equivalent to constancy of $r\sin\psi$.

Meridians vs. non-meridians. A meridian has $\theta' \equiv 0$, hence $c = 0$. A non-meridian geodesic has $c \neq 0$, so θ' never vanishes and the geodesic winds around the axis.

Problem 14. Visual exploration of minimal surfaces

Problem statement. Take a good look at pictures of minimal surfaces. If you have access to Mathematica (or another CAS/plotter), experiment with visualizing examples; alternatively, physical soap films can be used as intuition.

Solution. This is an exploratory problem rather than a proof problem. A good “solution” is to identify several distinct families of minimal surfaces, understand (qualitatively) what boundary conditions or symmetries produce them, and connect pictures to the theory:

- Minimal surfaces are critical points of the area functional (equivalently mean curvature $H \equiv 0$).
- Many classical examples admit Weierstrass representations and come in *associated families* (which preserve the induced metric while changing the embedding in $\mathbb{R}^3$).
- For surfaces produced as soap films, the boundary wire frame prescribes (approximately) a Plateau problem: the film is (ideally) a minimal surface spanning the given boundary.

Examples (minimal surfaces worth visualizing).

1. **Plane.** Trivial minimal surface ($H \equiv 0$).

2. **Catenoid.** Surface of revolution; classical soap film between two coaxial circles.
3. **Helicoid.** Ruled minimal surface; can be seen as an associated surface to the catenoid.
4. **Enneper surface.** Complete immersed minimal surface with self-intersections; simplest polynomial Weierstrass data.
5. **Scherk surfaces.** Doubly periodic examples; arise from boundary conditions resembling grids/planes.
6. **Costa surface.** A famous embedded example of finite total curvature with handles (genus).